Causal Analysis Report

BPI_Challenge_2013_closed_problems.xes.gz-dfg

Decision points:	8102
Choice sets:	6
Transitions fitted:	21
Significant (p<0.05):	1
Generated:	2026-06-09 10:23

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set. reject_H0=True → util predicts routing.

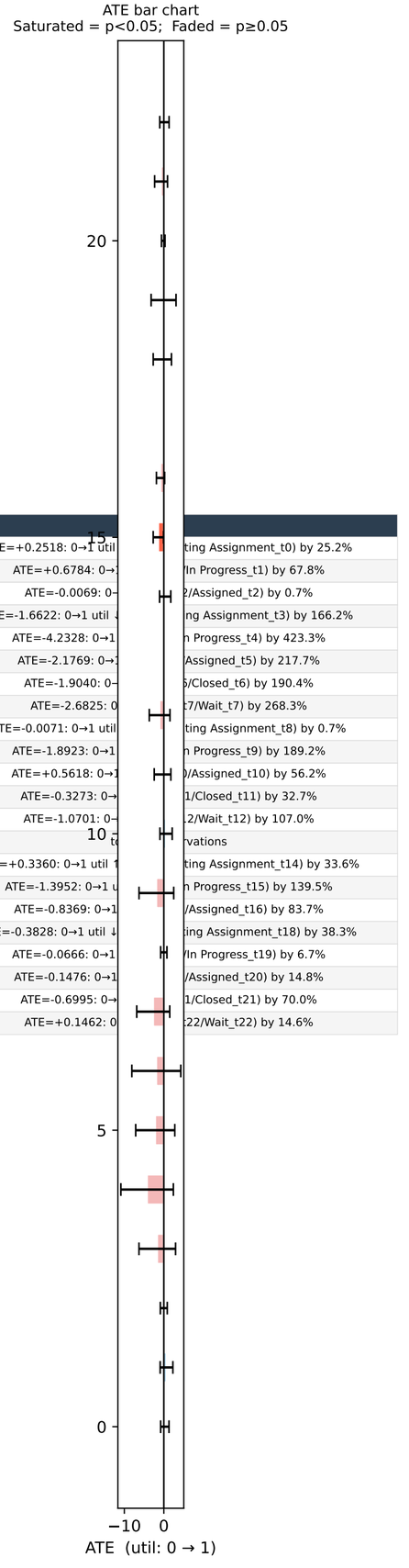
choice_set	choice_set_labels	n	choices_seen	coef	ci_95	perm_p_value	reject_H0	note
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	1.0724	[0.178, 1.987]	0.0035	True	higher util → t2 (Assigned_t2) over t0 (Awaiting Assignment_t0)
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	1.777	[1.338, 2.243]	0.0	True	higher util → t7 (Wait_t7) over t3 (Awaiting Assignment_t3)
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	-0.5348	[-1.264, 0.116]	0.214	False	higher util → t8 (Awaiting Assignment_t8) over t12 (Wait_t12)
(13, 17)	('In Progress_t13', 'ד_t17')	75	13	nan	—	nan	—	only t13 (In Progress_t13) ever chosen
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	0.7191	[0.131, 1.233]	0.08	False	higher util → t16 (Assigned_t16) over t14 (Awaiting Assignment_t14)
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	-0.3192	[-0.968, 0.321]	0.354	False	higher util → t18 (Awaiting Assignment_t18) over t22 (Wait_t22)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

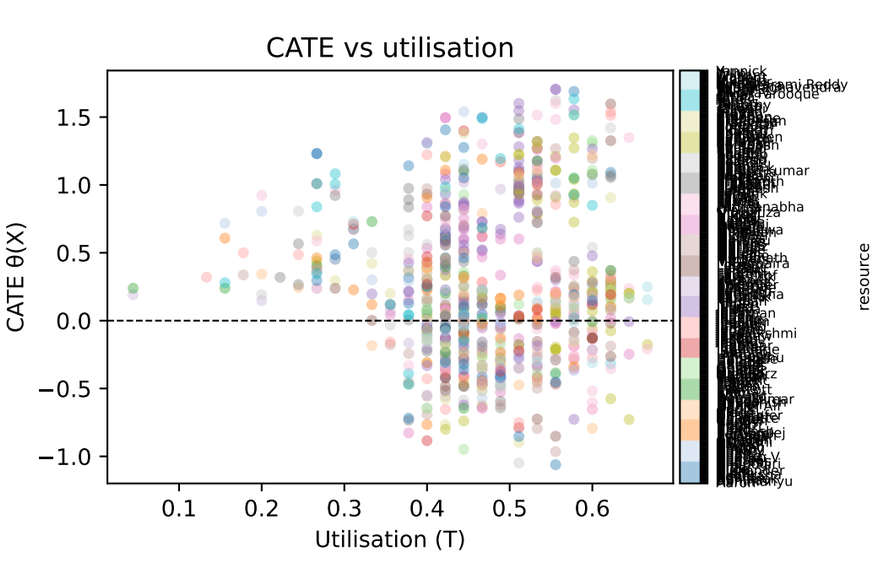
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	0	Awaiting Assignment_t0	Assigned_t2, In Progress_t1	0.48784	0.13932	[0.2148, 0.7609]	0.0005	True	θ=+0.4878: 1-unit ↑ util ↑ P(t0/Awaiting Assignment_t0) by 48.8%
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	1	In Progress_t1	Assigned_t2, Awaiting Assignment_t0	0.444249	0.250403	[-0.0465, 0.9350]	0.076	False	θ=+0.4442: 1-unit ↑ util ↑ P(t1/In Progress_t1) by 44.4%
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	2	Assigned_t2	Awaiting Assignment_t0, In Progress_t1	0.103218	0.130376	[-0.1523, 0.3588]	0.4285	False	θ=+0.1032: 1-unit ↑ util ↑ P(t2/Assigned_t2) by 10.3%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	3	Awaiting Assignment_t3	Assigned_t5, Closed_t6, In Progress_t4, Wait_t7	0.673421	0.234151	[0.2145, 1.1324]	0.004	True	θ=+0.6734: 1-unit ↑ util ↑ P(t3/Awaiting Assignment_t3) by 67.3%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	4	In Progress_t4	Assigned_t5, Awaiting Assignment_t3, Closed_t6, Wait_t7	0.338068	0.371865	[-0.3908, 1.0669]	0.3633	False	θ=+0.3381: 1-unit ↑ util ↑ P(t4/In Progress_t4) by 33.8%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	5	Assigned_t5	Awaiting Assignment_t3, Closed_t6, In Progress_t4, Wait_t7	-0.762486	0.315659	[-1.3812, -0.1438]	0.0157	True	θ=-0.7625: 1-unit ↑ util ↓ P(t5/Assigned_t5) by 76.2%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	6	Closed_t6	Assigned_t5, Awaiting Assignment_t3, In Progress_t4, Wait_t7	-0.900983	0.308404	[-1.5055, -0.2965]	0.0035	True	θ=-0.9010: 1-unit ↑ util ↓ P(t6/Closed_t6) by 90.1%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	7	Wait_t7	Assigned_t5, Awaiting Assignment_t3, Closed_t6, In Progress_t4	-0.786466	0.19695	[-1.1725, -0.4004]	0.0001	True	θ=-0.7865: 1-unit ↑ util ↓ P(t7/Wait_t7) by 78.6%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	8	Awaiting Assignment_t8	Assigned_t10, Closed_t11, In Progress_t9, Wait_t12	0.106346	0.137074	[-0.1623, 0.3750]	0.4379	False	θ=+0.1063: 1-unit ↑ util ↑ P(t8/Awaiting Assignment_t8) by 10.6%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	9	In Progress_t9	Assigned_t10, Awaiting Assignment_t8, Closed_t11, Wait_t12	0.915998	0.572789	[-0.2067, 2.0387]	0.1098	False	θ=+0.9160: 1-unit ↑ util ↑ P(t9/In Progress_t9) by 91.6%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	10	Assigned_t10	Awaiting Assignment_t8, Closed_t11, In Progress_t9, Wait_t12	-0.355499	0.227918	[-0.8022, 0.0912]	0.1188	False	θ=-0.3555: 1-unit ↑ util ↓ P(t10/Assigned_t10) by 35.5%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	11	Closed_t11	Assigned_t10, Awaiting Assignment_t8, In Progress_t9, Wait_t12	0.947741	0.316083	[0.3282, 1.5673]	0.0027	True	θ=+0.9477: 1-unit ↑ util ↑ P(t11/Closed_t11) by 94.8%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	12	Wait_t12	Assigned_t10, Awaiting Assignment_t8, Closed_t11, In Progress_t9	-0.249839	0.29639	[-0.8308, 0.3311]	0.3993	False	θ=-0.2498: 1-unit ↑ util ↓ P(t12/Wait_t12) by 25.0%
(13, 17)	('In Progress_t13', 'τ_t17')	75	13	—	—	—	nan	nan	—	nan	—	too few observations
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	14	Awaiting Assignment_t14	Assigned_t16, In Progress_t15	-0.037513	0.429987	[-0.8803, 0.8053]	0.9305	False	θ=-0.0375: 1-unit ↑ util ↓ P(t14/Awaiting Assignment_t14) by 3.8%
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	15	In Progress_t15	Assigned_t16, Awaiting Assignment_t14	-1.029833	0.418589	[-1.8503, -0.2094]	0.0139	True	θ=-1.0298: 1-unit ↑ util ↓ P(t15/In Progress_t15) by 103.0%
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	16	Assigned_t16	Awaiting Assignment_t14, In Progress_t15	-0.643511	0.320412	[-1.2715, -0.0155]	0.0446	True	θ=-0.6435: 1-unit ↑ util ↓ P(t16/Assigned_t16) by 64.4%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	18	Awaiting Assignment_t18	Assigned_t20, Closed_t21, In Progress_t19, Wait_t22	-0.185693	0.15373	[-0.4870, 0.1156]	0.2271	False	θ=-0.1857: 1-unit ↑ util ↓ P(t18/Awaiting Assignment_t18) by 18.6%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	19	In Progress_t19	Assigned_t20, Awaiting Assignment_t18, Closed_t21, Wait_t22	0.101827	0.181823	[-0.2545, 0.4582]	0.5755	False	θ=+0.1018: 1-unit ↑ util ↑ P(t19/In Progress_t19) by 10.2%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	20	Assigned_t20	Awaiting Assignment_t18, Closed_t21, In Progress_t19, Wait_t22	-0.08543	0.042215	[-0.1682, -0.0027]	0.043	True	θ=-0.0854: 1-unit ↑ util ↓ P(t20/Assigned_t20) by 8.5%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	21	Closed_t21	Assigned_t20, Awaiting Assignment_t18, In Progress_t19, Wait_t22	-0.000533	0.015981	[-0.0319, 0.0308]	0.9734	False	θ=-0.0005: 1-unit ↑ util ↓ P(t21/Closed_t21) by 0.1%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	22	Wait_t22	Assigned_t20, Awaiting Assignment_t18, Closed_t21, In Progress_t19	0.052389	0.037485	[-0.0211, 0.1259]	0.1622	False	θ=+0.0524: 1-unit ↑ util ↑ P(t22/Wait_t22) by 5.2%

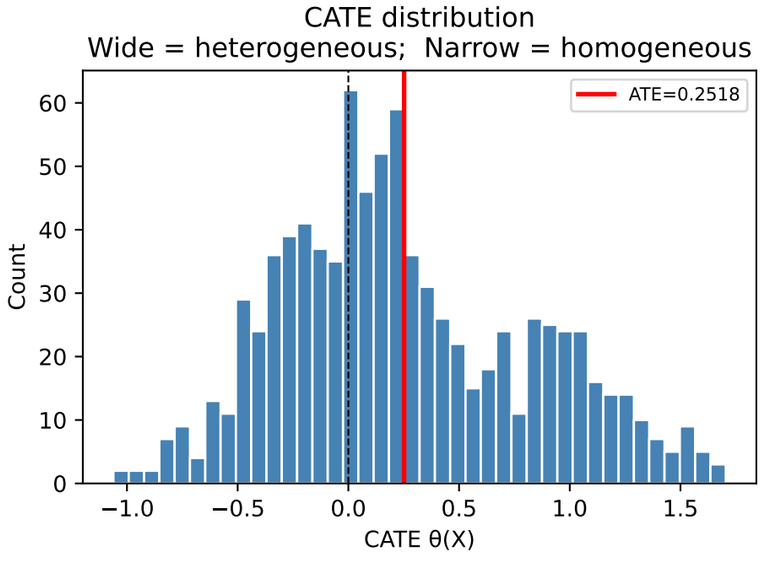
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

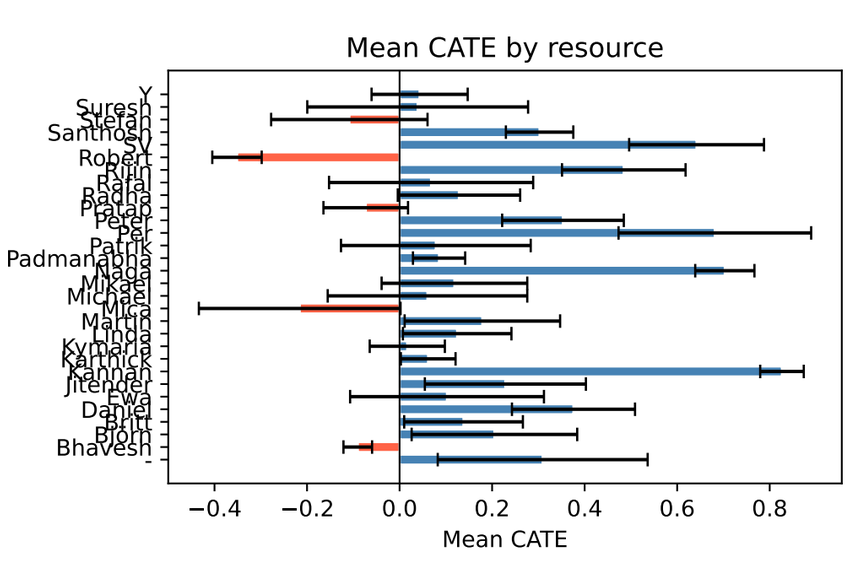
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	0	Awaiting Assignment_t0	Assigned_t2, In Progress_t1	0.251788	[-0.8023, 1.3058]	0.6396	False	ATE=+0.2518: 0→1 util ting Assignment_t0) by 25.2%
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	1	In Progress_t1	Assigned_t2, Awaiting Assignment_t0	0.678355	[-0.9320, 2.2887]	0.409	False	ATE=+0.6784: 0→ In Progress_t1) by 67.8%
(0, 1, 2)	('Awaiting Assignment_t0', 'In Progress_t1', 'Assigned_t2')	875	0, 1, 2	2	Assigned_t2	Awaiting Assignment_t0, In Progress_t1	-0.006889	[-0.8869, 0.8731]	0.9878	False	ATE=-0.0069: 0– 2/Assigned_t2) by 0.7%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	3	Awaiting Assignment_t3	Assigned_t5, Closed_t6, In Progress_t4, Wait_t7	-1.662155	[-6.2932, 2.9688]	0.4818	False	ATE=-1.6622: 0→1 util ng Assignment_t3) by 166.2%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	4	In Progress_t4	Assigned_t5, Awaiting Assignment_t3, Closed_t6, Wait_t7	-4.232796	[-10.8763, 2.4107]	0.2117	False	ATE=-4.2328: 0→1 n Progress_t4) by 423.3%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	5	Assigned_t5	Awaiting Assignment_t3, Closed_t6, In Progress_t4, Wait_t7	-2.176946	[-7.1244, 2.7705]	0.3884	False	ATE=-2.1769: 0→ Assigned_t5) by 217.7%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	6	Closed_t6	Assigned_t5, Awaiting Assignment_t3, In Progress_t4, Wait_t7	-1.904017	[-8.0990, 4.2909]	0.5469	False	ATE=-1.9040: 0– 6/Closed_t6) by 190.4%
(3, 4, 5, 6, 7)	('Awaiting Assignment_t3', 'In Progress_t4', 'Assigned_t5', 'Closed_t6', 'Wait_t7')	3101	3, 4, 5, 6, 7	7	Wait_t7	Assigned_t5, Awaiting Assignment_t3, Closed_t6, In Progress_t4	-2.682526	[-6.8590, 1.4939]	0.2081	False	ATE=-2.6825: 0 t7/Wait_t7) by 268.3%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	8	Awaiting Assignment_t8	Assigned_t10, Closed_t11, In Progress_t9, Wait_t12	-0.007146	[-0.7992, 0.7850]	0.9859	False	ATE=-0.0071: 0→1 util ting Assignment_t8) by 0.7%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	9	In Progress_t9	Assigned_t10, Awaiting Assignment_t8, Closed_t11, Wait_t12	-1.892258	[-6.2846, 2.5001]	0.3985	False	ATE=-1.8923: 0→1 n Progress_t9) by 189.2%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	10	Assigned_t10	Awaiting Assignment_t8, Closed_t11, In Progress_t9, Wait_t12	0.561794	[-0.9930, 2.1166]	0.4788	False	ATE=+0.5618: 0→1 0/Assigned_t10) by 56.2%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	11	Closed_t11	Assigned_t10, Awaiting Assignment_t8, In Progress_t9, Wait_t12	-0.327334	[-2.4352, 1.7805]	0.7608	False	ATE=-0.3273: 0→ 1/Closed_t11) by 32.7%
(8, 9, 10, 11, 12)	('Awaiting Assignment_t8', 'In Progress_t9', 'Assigned_t10', 'Closed_t11', 'Wait_t12')	704	8, 9, 10, 11, 12	12	Wait_t12	Assigned_t10, Awaiting Assignment_t8, Closed_t11, In Progress_t9	-1.070115	[-3.6889, 1.5487]	0.4232	False	ATE=-1.0701: 0– 2/Wait_t12) by 107.0%
(13, 17)	('In Progress_t13', 'τ_t17')	75	13	—	—	—	nan	—	nan	—	t vations
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	14	Awaiting Assignment_t14	Assigned_t16, In Progress_t15	0.335976	[-1.0840, 1.7559]	0.6428	False	ATE=+0.3360: 0→1 util ting Assignment_t14) by 33.6%
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	15	In Progress_t15	Assigned_t16, Awaiting Assignment_t14	-1.395221	[-2.6929, -0.0975]	0.0351	True	ATE=-1.3952: 0→1 u n Progress_t15) by 139.5%
(14, 15, 16)	('Awaiting Assignment_t14', 'In Progress_t15', 'Assigned_t16')	248	14, 15, 16	16	Assigned_t16	Awaiting Assignment_t14, In Progress_t15	-0.836938	[-1.8804, 0.2065]	0.1159	False	ATE=-0.8369: 0→1 /Assigned_t16) by 83.7%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	18	Awaiting Assignment_t18	Assigned_t20, Closed_t21, In Progress_t19, Wait_t22	-0.382844	[-2.6916, 1.9259]	0.7452	False	ATE=-0.3828: 0→1 util ing Assignment_t18) by 38.3%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	19	In Progress_t19	Assigned_t20, Awaiting Assignment_t18, Closed_t21, Wait_t22	-0.066567	[-3.2193, 3.0862]	0.967	False	ATE=-0.0666: 0→1 /In Progress_t19) by 6.7%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	20	Assigned_t20	Awaiting Assignment_t18, Closed_t21, In Progress_t19, Wait_t22	-0.147636	[-0.6008, 0.3056]	0.5231	False	ATE=-0.1476: 0→1 /Assigned_t20) by 14.8%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	21	Closed_t21	Assigned_t20, Awaiting Assignment_t18, In Progress_t19, Wait_t22	-0.699531	[-2.3347, 0.9357]	0.4018	False	ATE=-0.6995: 0→ 1/Closed_t21) by 70.0%
(18, 19, 20, 21, 22)	('Awaiting Assignment_t18', 'In Progress_t19', 'Assigned_t20', 'Closed_t21', 'Wait_t22')	1487	18, 19, 20, 21, 22	22	Wait_t22	Assigned_t20, Awaiting Assignment_t18, Closed_t21, In Progress_t19	0.146208	[-1.0358, 1.3282]	0.8084	False	ATE=+0.1462: 0 22/Wait_t22) by 14.6%

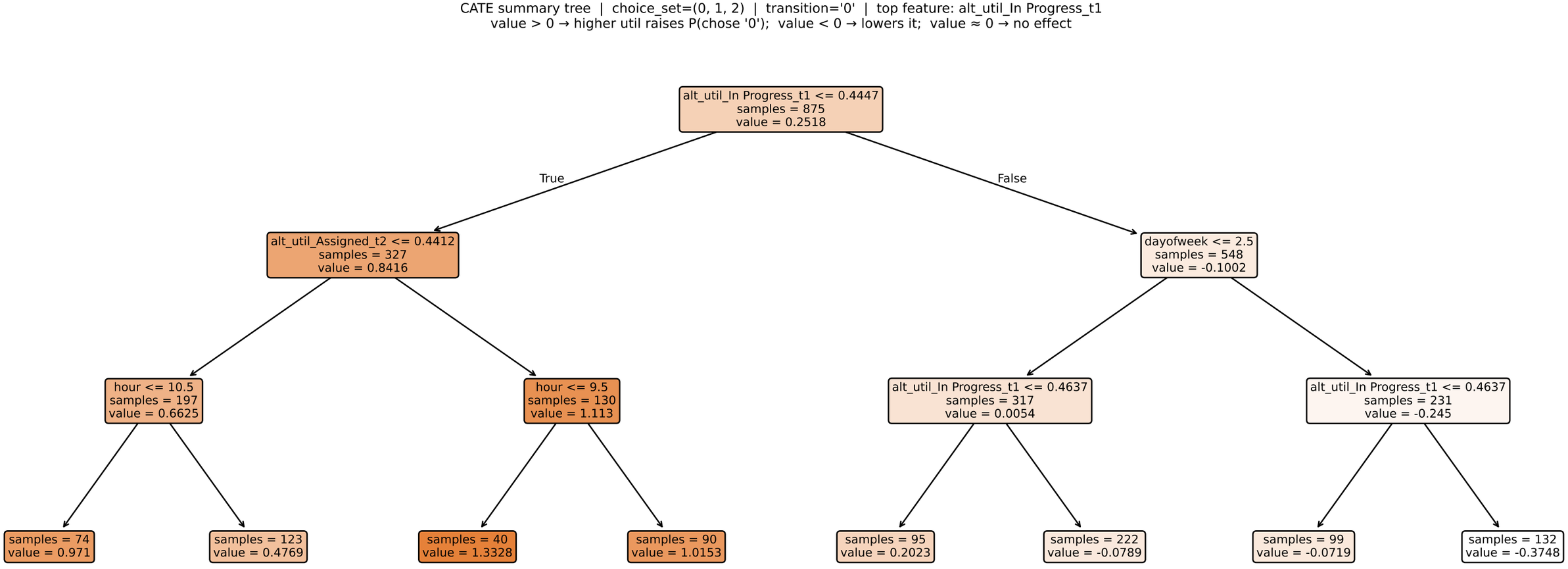


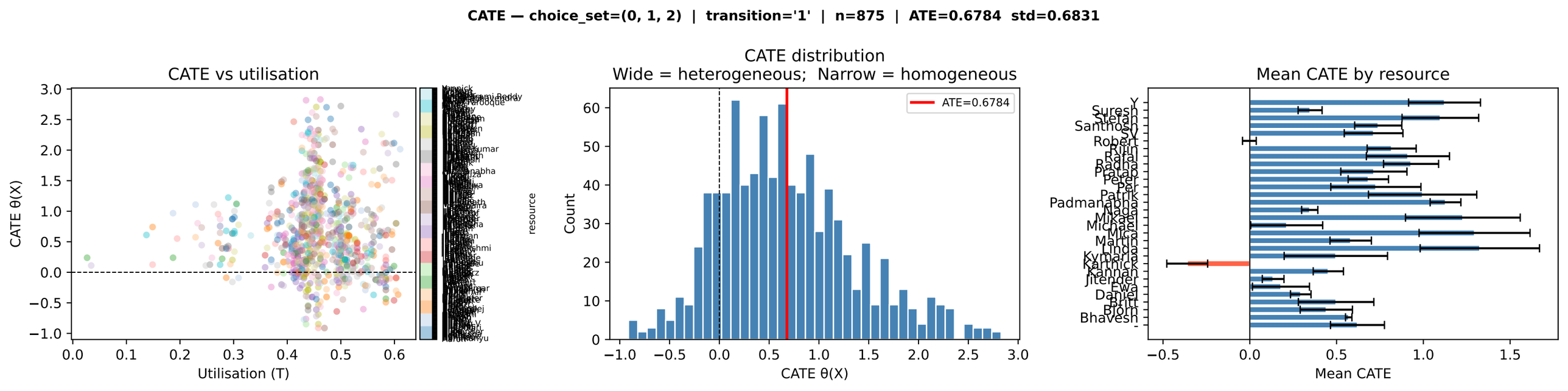
CATE — choice_set=(0, 1, 2) | transition='0' | n=875 | ATE=0.2518 std=0.5625

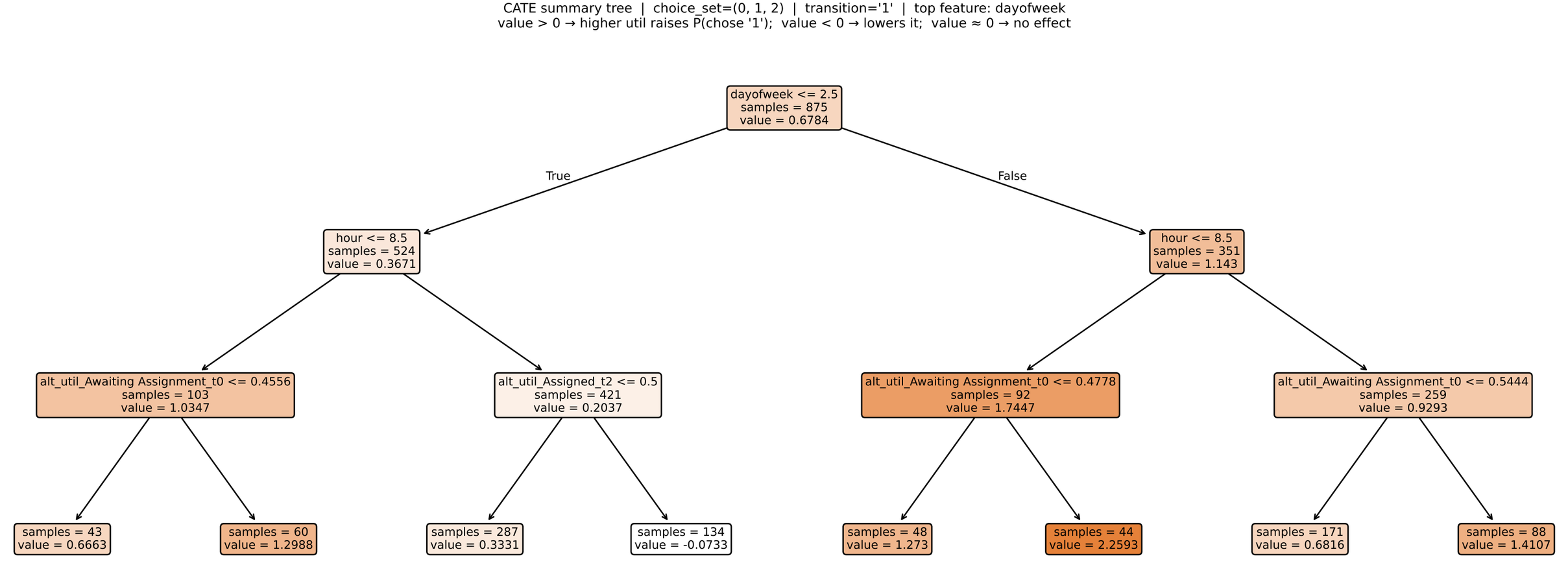


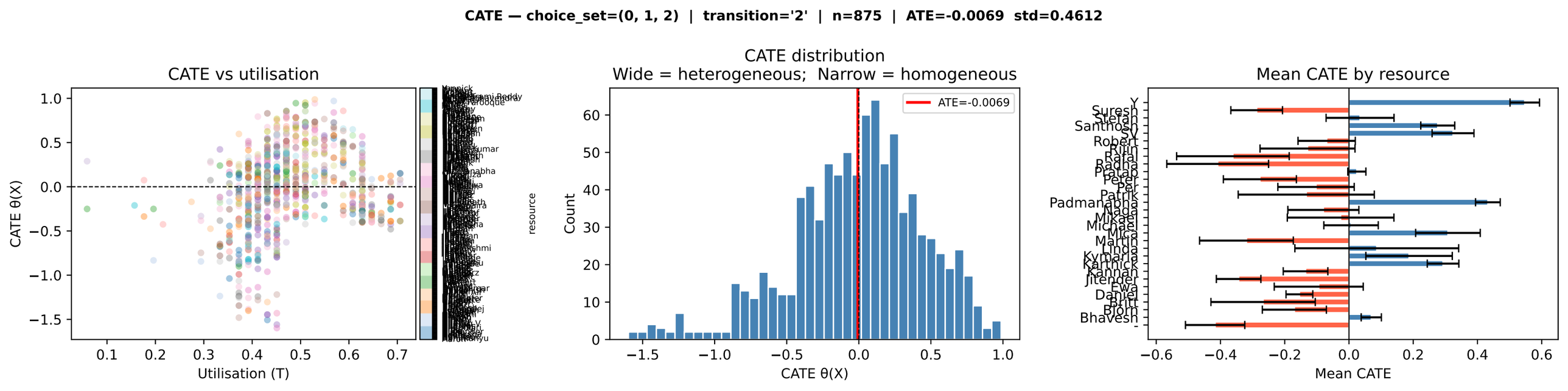


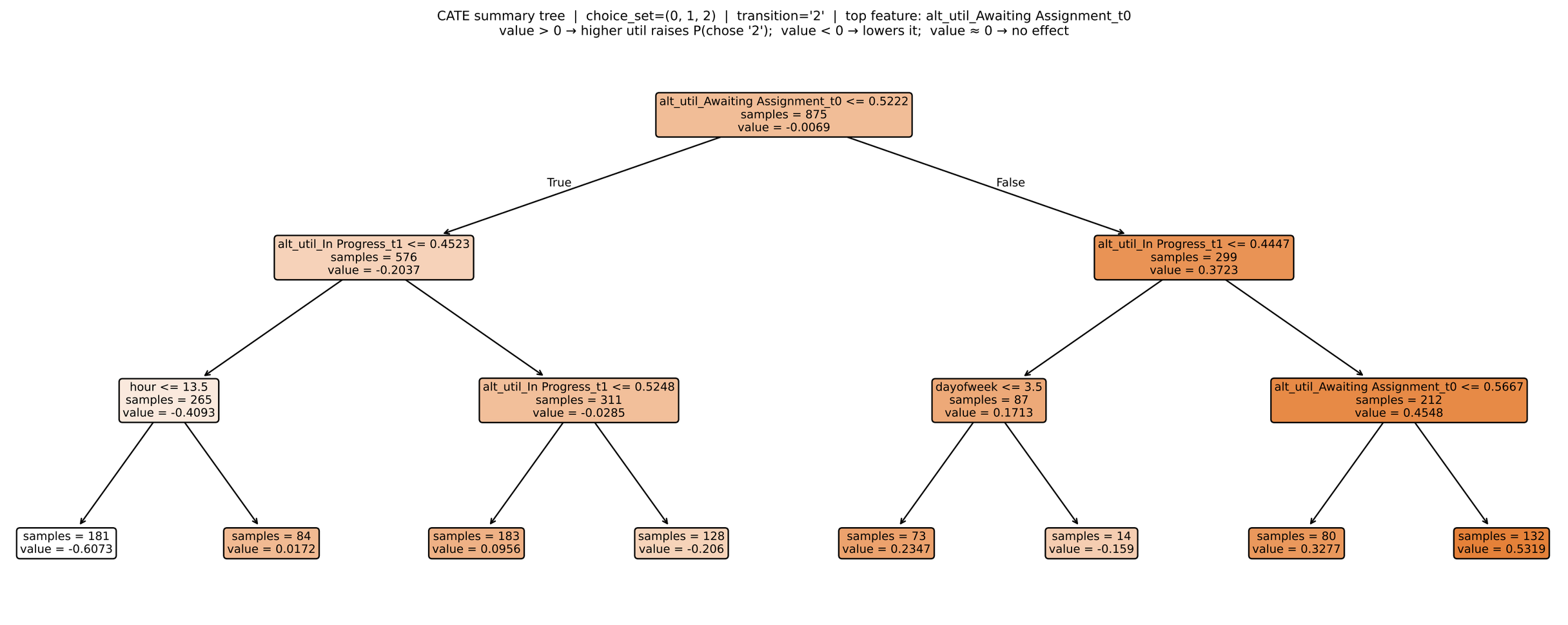


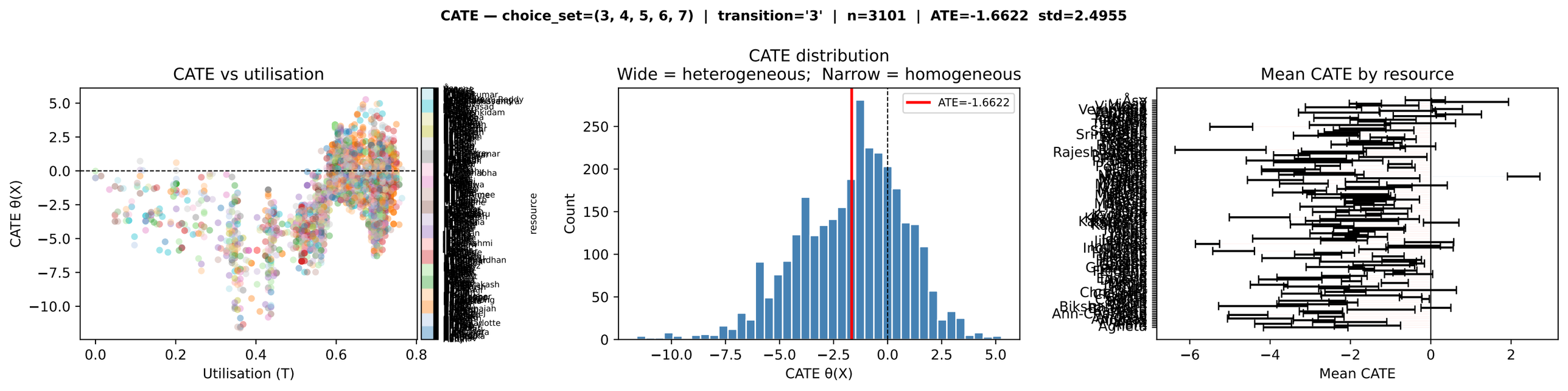

CATE summary tree | choice_set=(0, 1, 2) | transition='0' | top feature: alt_util_ln Progress_t1
value > 0 → higher util raises P(chose '0'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t1 <= 0.4447
samples = 875
value = 0.2518
True
False
alt_util_Assigned_t2 <= 0.4412
samples = 327
value = 0.8416
dayofweek <= 2.5
samples = 548
value = -0.1002
hour <= 10.5
samples = 197
value = 0.6625
hour <= 9.5
samples = 130
value = 1.113
alt_util_ln Progress_t1 <= 0.4637
samples = 317
value = 0.0054
alt_util_ln Progress_t1 <= 0.4637
samples = 231
value = -0.245
samples = 74
value = 0.971
samples = 123
value = 0.4769
samples = 40
value = 1.3328
samples = 90
value = 1.0153
samples = 95
value = 0.2023
samples = 222
value = -0.0789
samples = 99
value = -0.0719
samples = 132
value = -0.3748

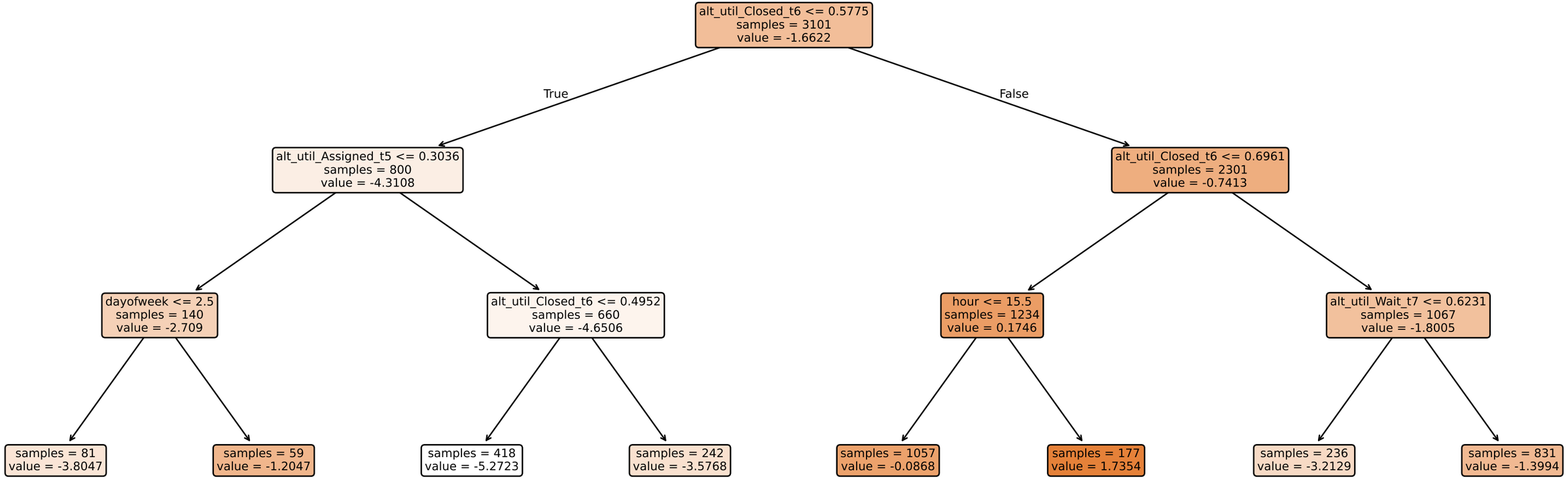

CATE — choice_set=(0, 1, 2) | transition='1' | n=875 | ATE=0.6784 std=0.6831
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.6784
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(0, 1, 2) | transition='1' | top feature: dayofweek
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 875
value = 0.6784
True
False
hour <= 8.5
samples = 524
value = 0.3671
hour <= 8.5
samples = 351
value = 1.143
alt_util_Awaiting Assignment_t0 <= 0.4556
samples = 103
value = 1.0347
alt_util_Assigned_t2 <= 0.5
samples = 421
value = 0.2037
alt_util_Awaiting Assignment_t0 <= 0.4778
samples = 92
value = 1.7447
alt_util_Awaiting Assignment_t0 <= 0.5444
samples = 259
value = 0.9293
samples = 43
value = 0.6663
samples = 60
value = 1.2988
samples = 287
value = 0.3331
samples = 134
value = -0.0733
samples = 48
value = 1.273
samples = 44
value = 2.2593
samples = 171
value = 0.6816
samples = 88
value = 1.4107

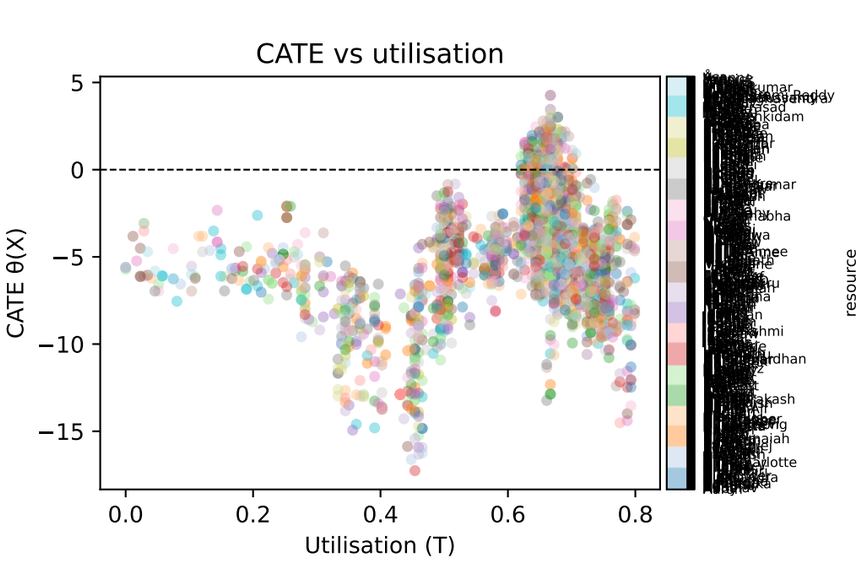

CATE — choice_set=(0, 1, 2) | transition='2' | n=875 | ATE=-0.0069 std=0.4612
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0069
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

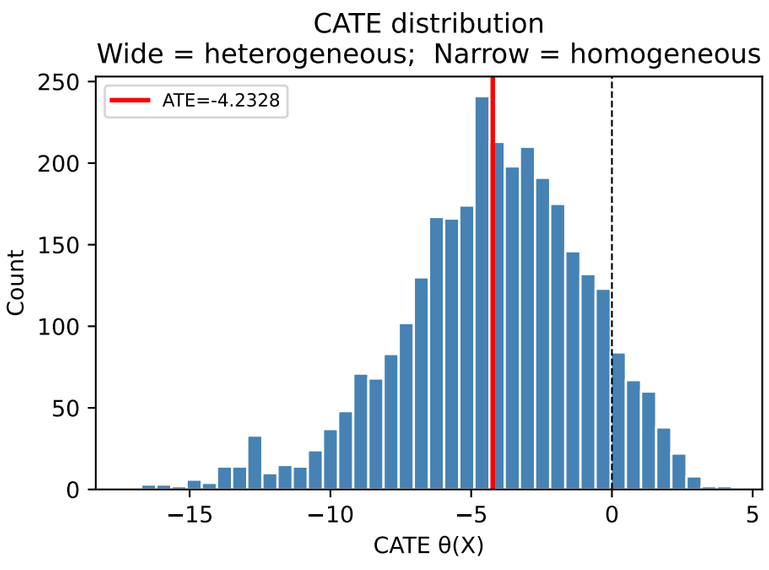

CATE summary tree | choice_set=(0, 1, 2) | transition='2' | top feature: alt_util_Awaiting Assignment_t0
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t0 <= 0.5222
samples = 875
value = -0.0069
True
False
alt_util_In Progress_t1 <= 0.4523
samples = 576
value = -0.2037
alt_util_In Progress_t1 <= 0.4447
samples = 299
value = 0.3723
hour <= 13.5
samples = 265
value = -0.4093
alt_util_In Progress_t1 <= 0.5248
samples = 311
value = -0.0285
dayofweek <= 3.5
samples = 87
value = 0.1713
alt_util_Awaiting Assignment_t0 <= 0.5667
samples = 212
value = 0.4548
samples = 181
value = -0.6073
samples = 84
value = 0.0172
samples = 183
value = 0.0956
samples = 128
value = -0.206
samples = 73
value = 0.2347
samples = 14
value = -0.159
samples = 80
value = 0.3277
samples = 132
value = 0.5319

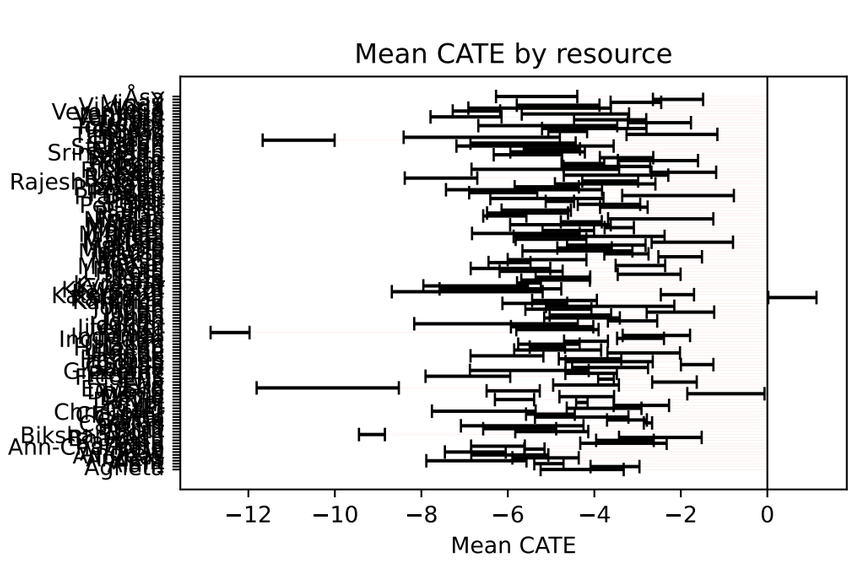

CATE — choice_set=(3, 4, 5, 6, 7) | transition='3' | n=3101 | ATE=-1.6622 std=2.4955
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.6622
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

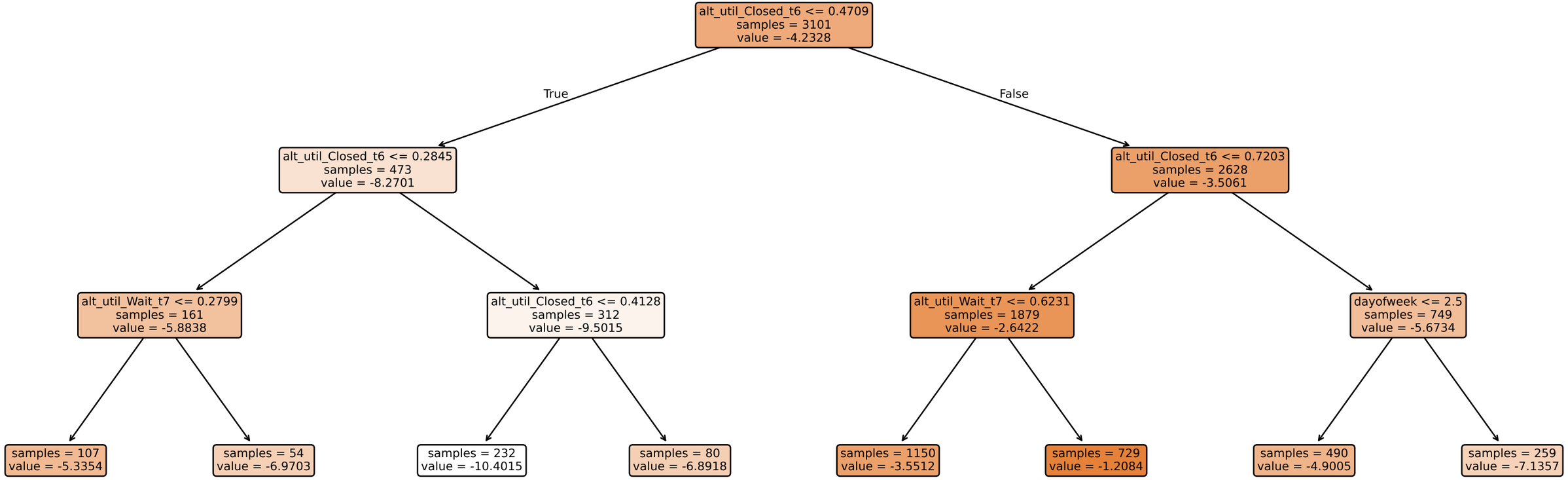

CATE summary tree | choice_set=(3, 4, 5, 6, 7) | transition='3' | top feature: alt_util_Closed_t6
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t6 <= 0.5775
samples = 3101
value = -1.6622
True
False
alt_util_Assigned_t5 <= 0.3036
samples = 800
value = -4.3108
alt_util_Closed_t6 <= 0.6961
samples = 2301
value = -0.7413
dayofweek <= 2.5
samples = 140
value = -2.709
alt_util_Closed_t6 <= 0.4952
samples = 660
value = -4.6506
hour <= 15.5
samples = 1234
value = 0.1746
alt_util_Wait_t7 <= 0.6231
samples = 1067
value = -1.8005
samples = 81
value = -3.8047
samples = 59
value = -1.2047
samples = 418
value = -5.2723
samples = 242
value = -3.5768
samples = 1057
value = -0.0868
samples = 177
value = 1.7354
samples = 236
value = -3.2129
samples = 831
value = -1.3994

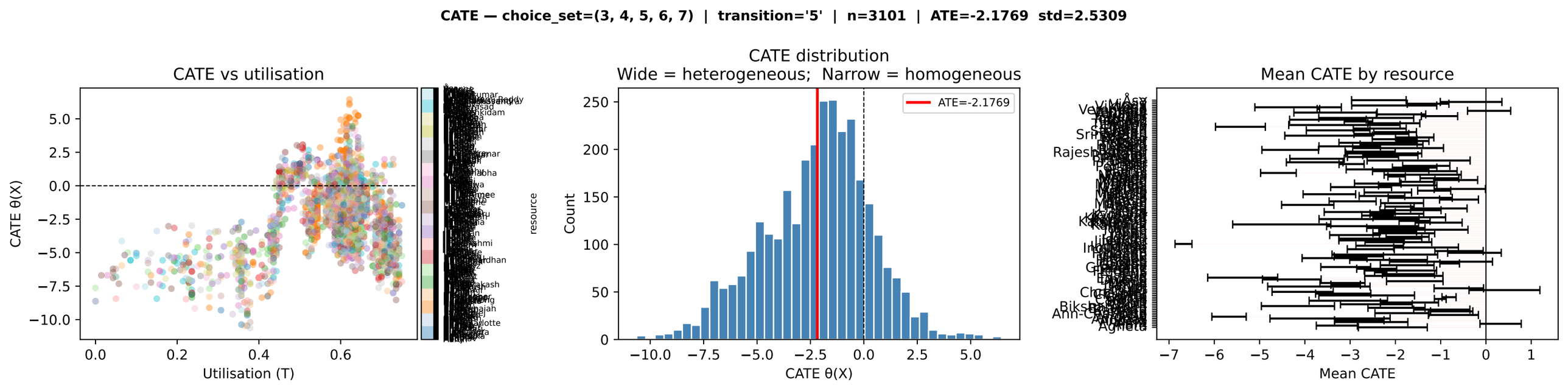
CATE — choice_set=(3, 4, 5, 6, 7) | transition='4' | n=3101 | ATE=-4.2328 std=3.3212

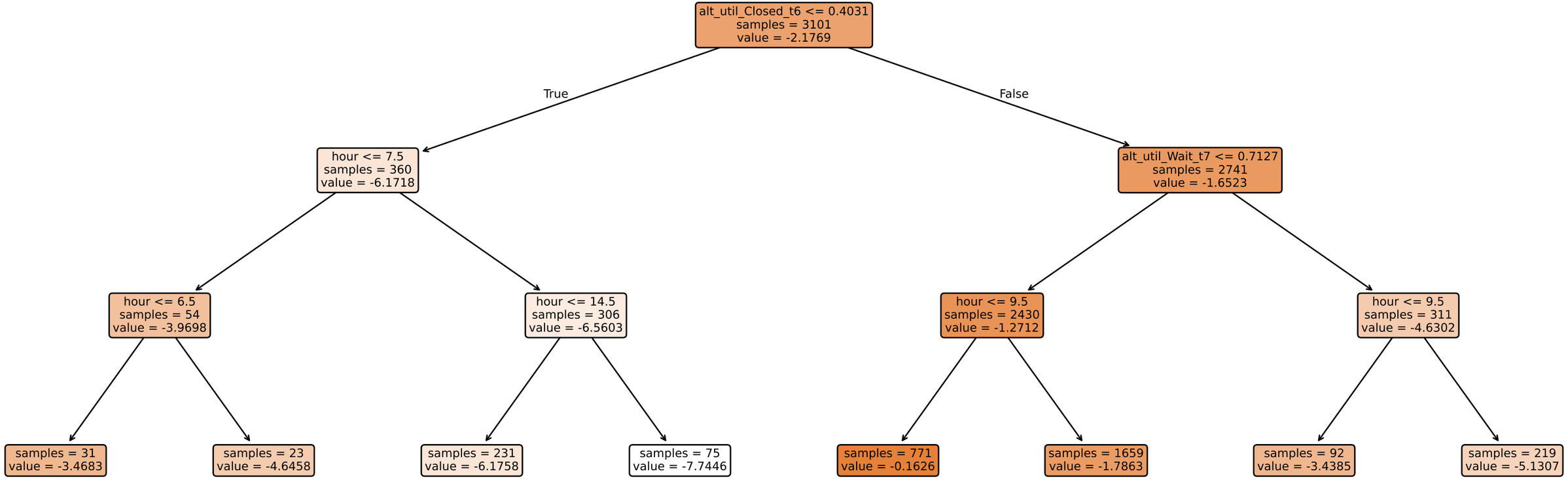


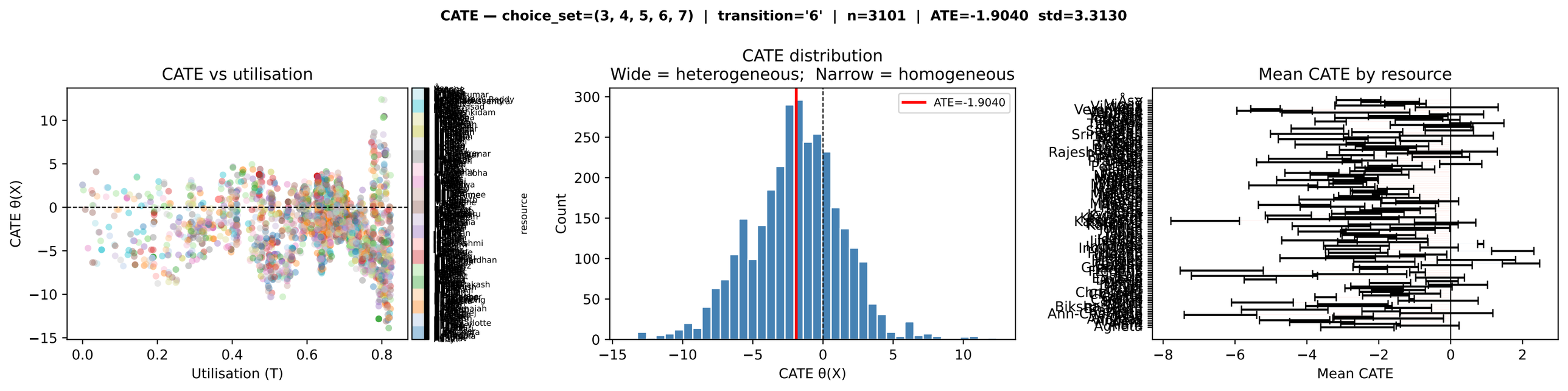


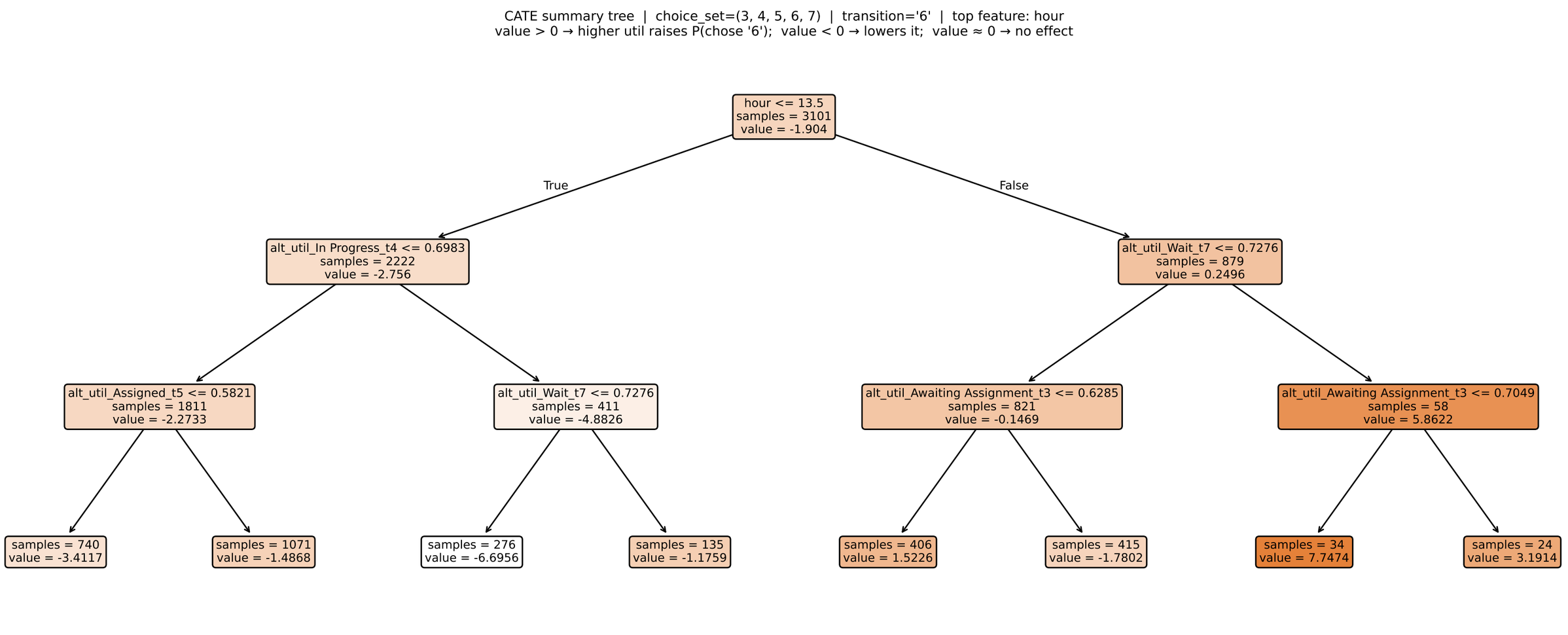


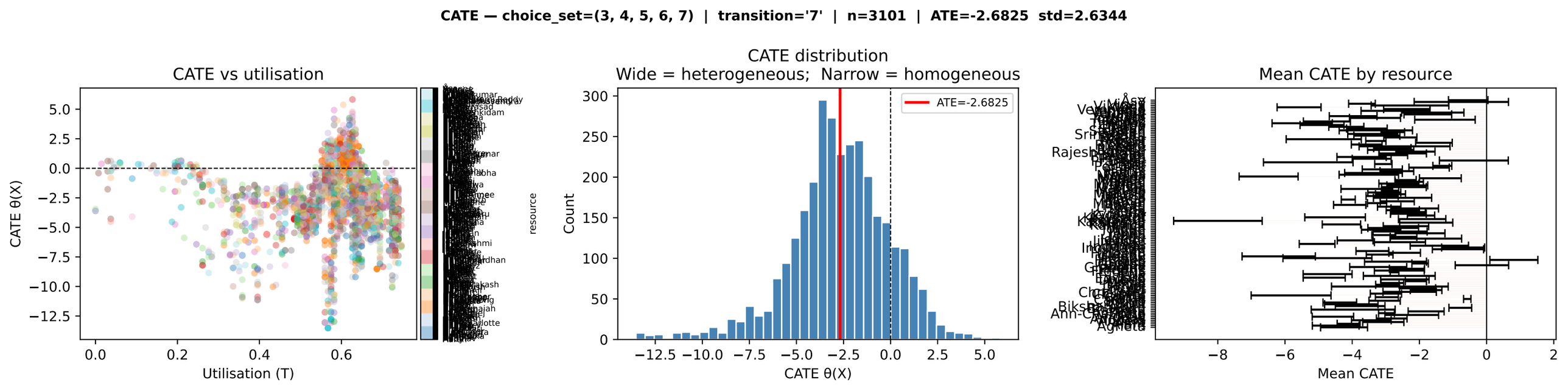

CATE summary tree | choice_set=(3, 4, 5, 6, 7) | transition='4' | top feature: alt_util_Closed_t6
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t6 <= 0.4709
samples = 3101
value = -4.2328
True
False
alt_util_Closed_t6 <= 0.2845
samples = 473
value = -8.2701
alt_util_Closed_t6 <= 0.7203
samples = 2628
value = -3.5061
alt_util_Wait_t7 <= 0.2799
samples = 161
value = -5.8838
alt_util_Closed_t6 <= 0.4128
samples = 312
value = -9.5015
alt_util_Wait_t7 <= 0.6231
samples = 1879
value = -2.6422
dayofweek <= 2.5
samples = 749
value = -5.6734
samples = 107
value = -5.3354
samples = 54
value = -6.9703
samples = 232
value = -10.4015
samples = 80
value = -6.8918
samples = 1150
value = -3.5512
samples = 729
value = -1.2084
samples = 490
value = -4.9005
samples = 259
value = -7.1357

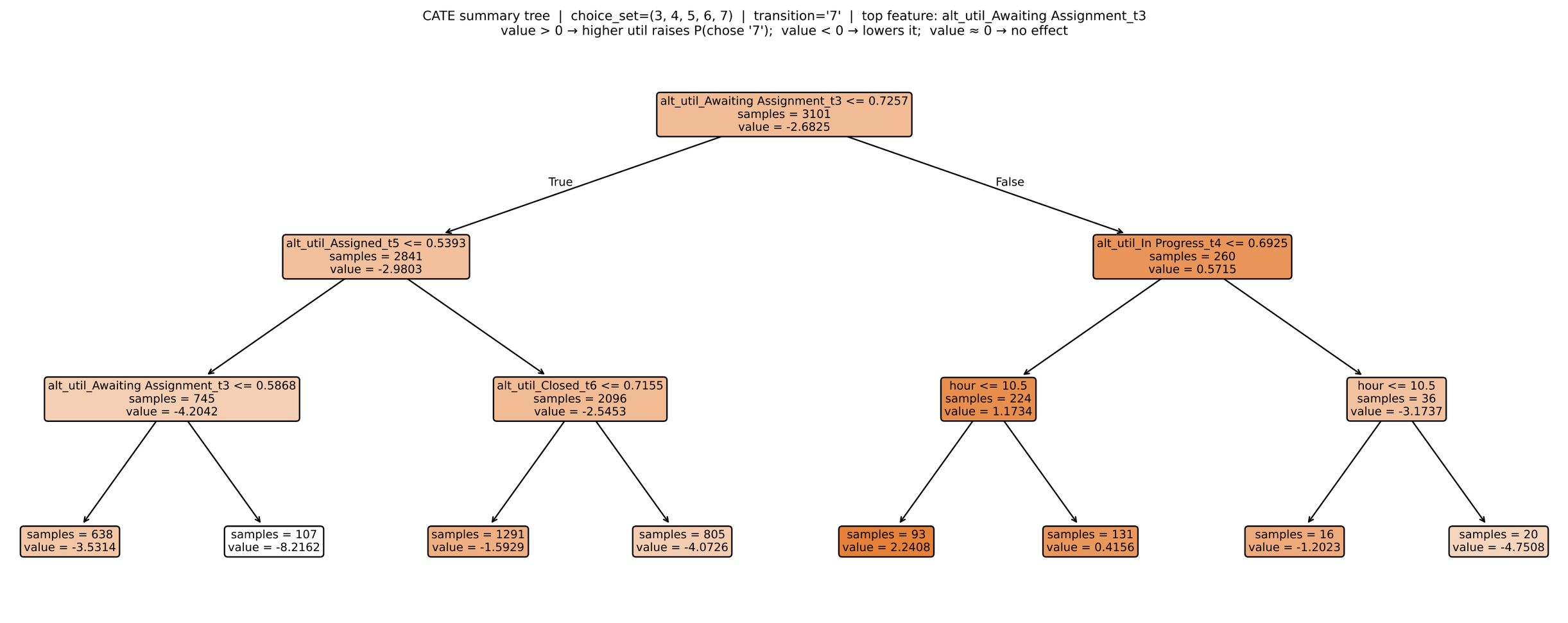

CATE — choice_set=(3, 4, 5, 6, 7) | transition='5' | n=3101 | ATE=-2.1769 std=2.5309
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.1769
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

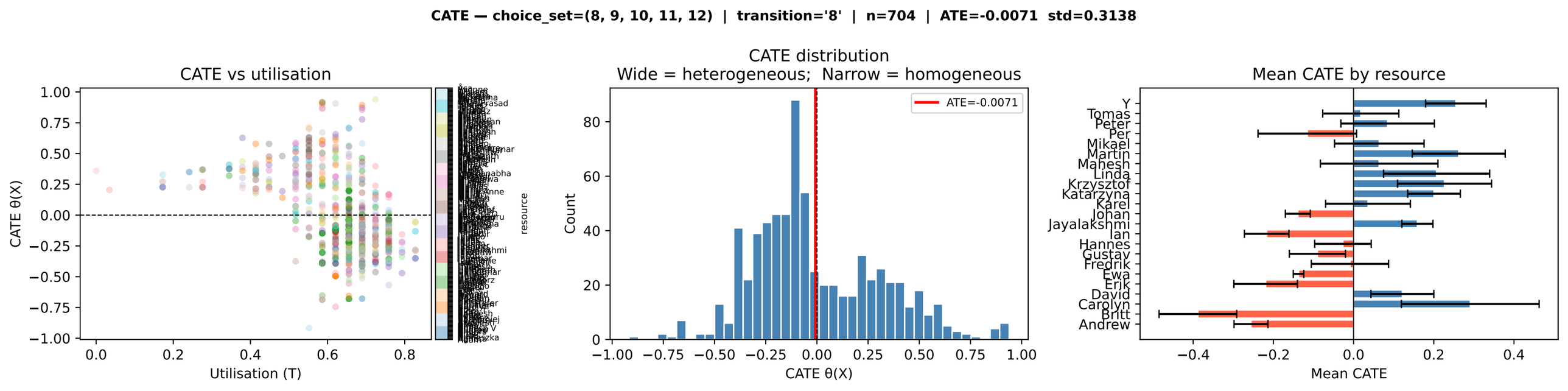

CATE summary tree | choice_set=(3, 4, 5, 6, 7) | transition='5' | top feature: alt_util_Closed_t6
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t6 <= 0.4031
samples = 3101
value = -2.1769
True
False
hour <= 7.5
samples = 360
value = -6.1718
alt_util_Wait_t7 <= 0.7127
samples = 2741
value = -1.6523
hour <= 6.5
samples = 54
value = -3.9698
hour <= 14.5
samples = 306
value = -6.5603
hour <= 9.5
samples = 2430
value = -1.2712
hour <= 9.5
samples = 311
value = -4.6302
samples = 31
value = -3.4683
samples = 23
value = -4.6458
samples = 231
value = -6.1758
samples = 75
value = -7.7446
samples = 771
value = -0.1626
samples = 1659
value = -1.7863
samples = 92
value = -3.4385
samples = 219
value = -5.1307

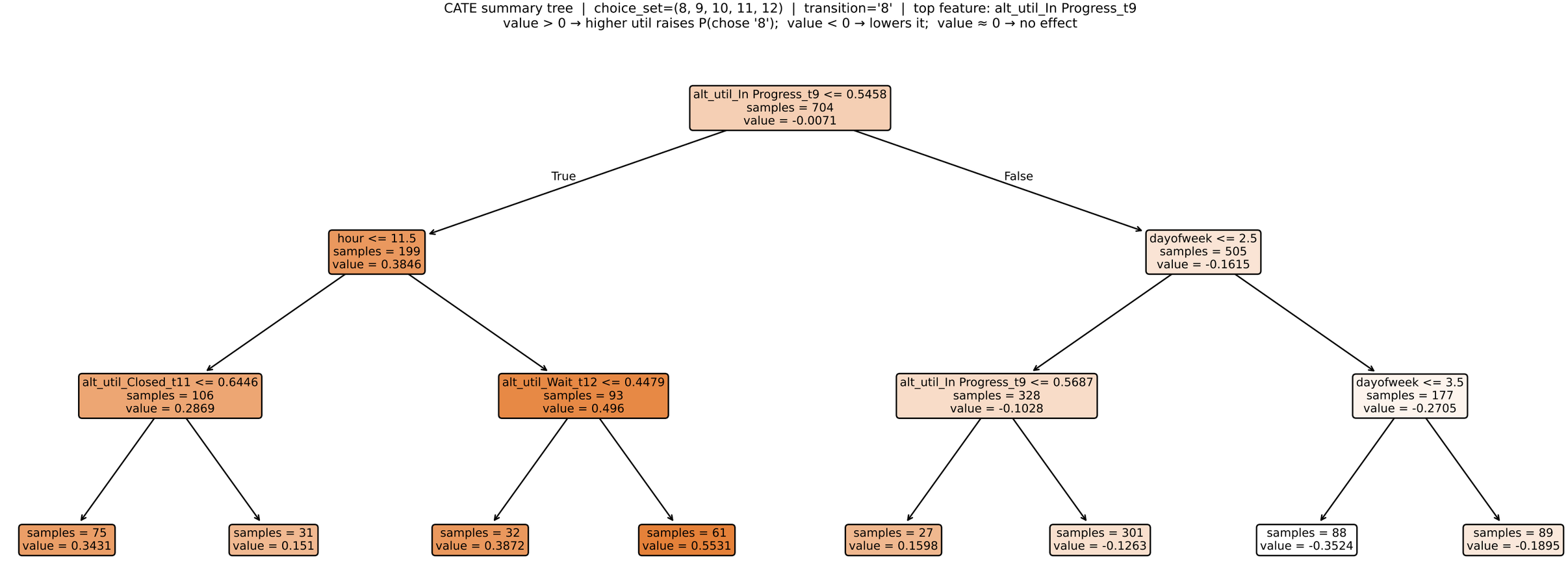

CATE — choice_set=(3, 4, 5, 6, 7) | transition='6' | n=3101 | ATE=-1.9040 std=3.3130
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.9040
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

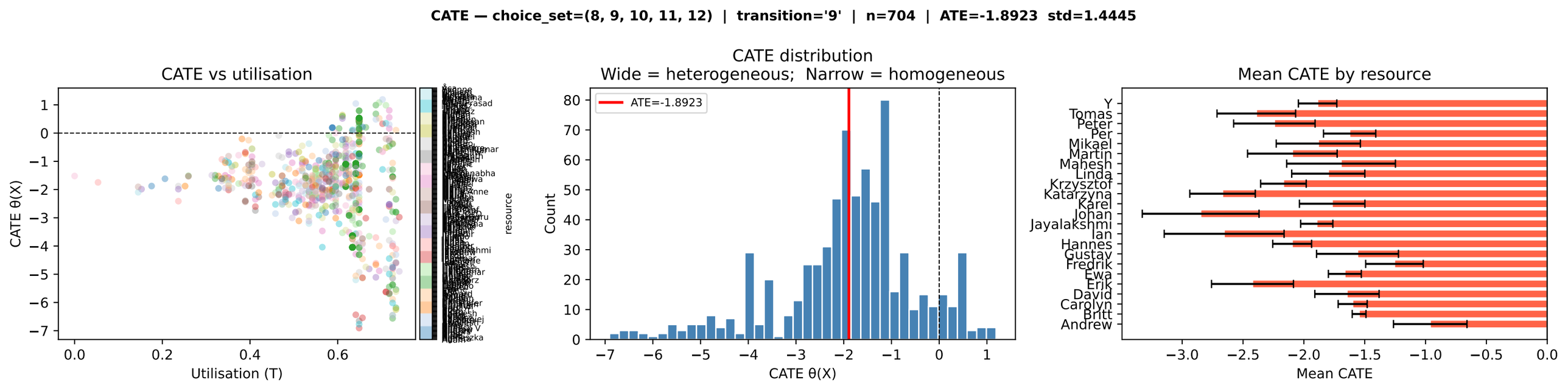

CATE summary tree | choice_set=(3, 4, 5, 6, 7) | transition='6' | top feature: hour
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 3101
value = -1.904
True
False
alt_util_In Progress_t4 <= 0.6983
samples = 2222
value = -2.756
alt_util_Wait_t7 <= 0.7276
samples = 879
value = 0.2496
alt_util_Assigned_t5 <= 0.5821
samples = 1811
value = -2.2733
alt_util_Wait_t7 <= 0.7276
samples = 411
value = -4.8826
alt_util_Awaiting Assignment_t3 <= 0.6285
samples = 821
value = -0.1469
alt_util_Awaiting Assignment_t3 <= 0.7049
samples = 58
value = 5.8622
samples = 740
value = -3.4117
samples = 1071
value = -1.4868
samples = 276
value = -6.6956
samples = 135
value = -1.1759
samples = 406
value = 1.5226
samples = 415
value = -1.7802
samples = 34
value = 7.7474
samples = 24
value = 3.1914

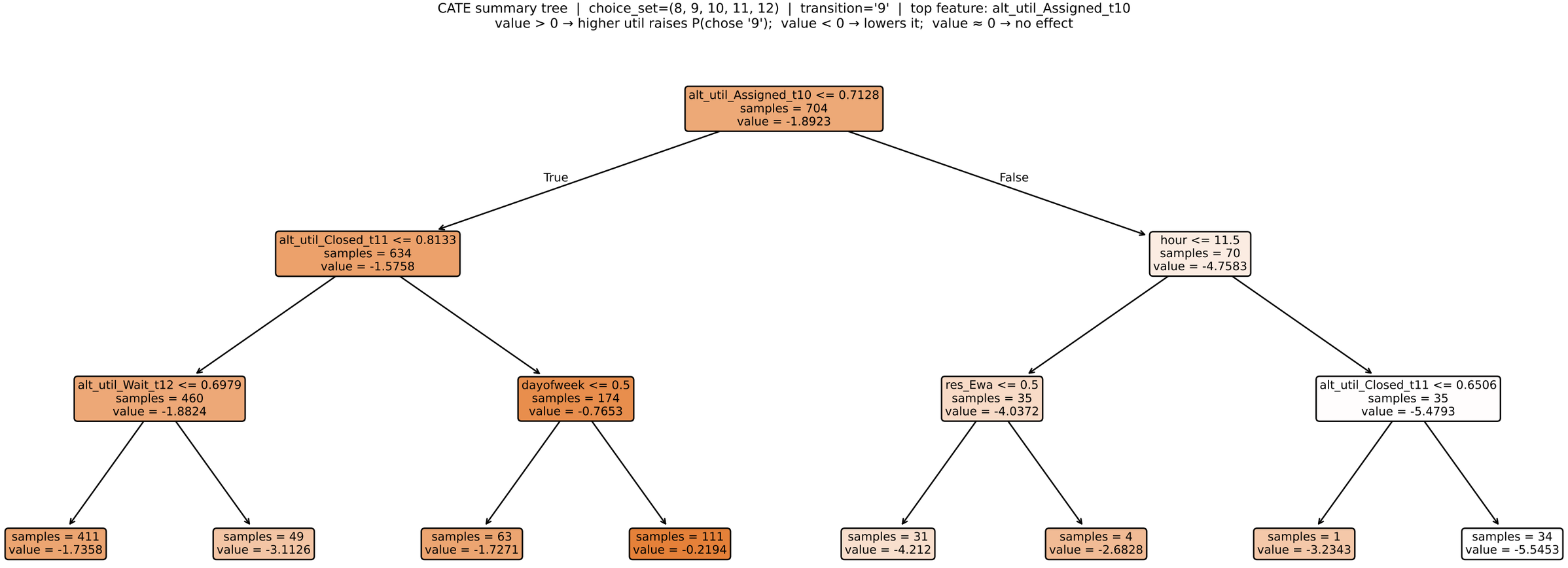

CATE — choice_set=(3, 4, 5, 6, 7) | transition='7' | n=3101 | ATE=-2.6825 std=2.6344
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.6825
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

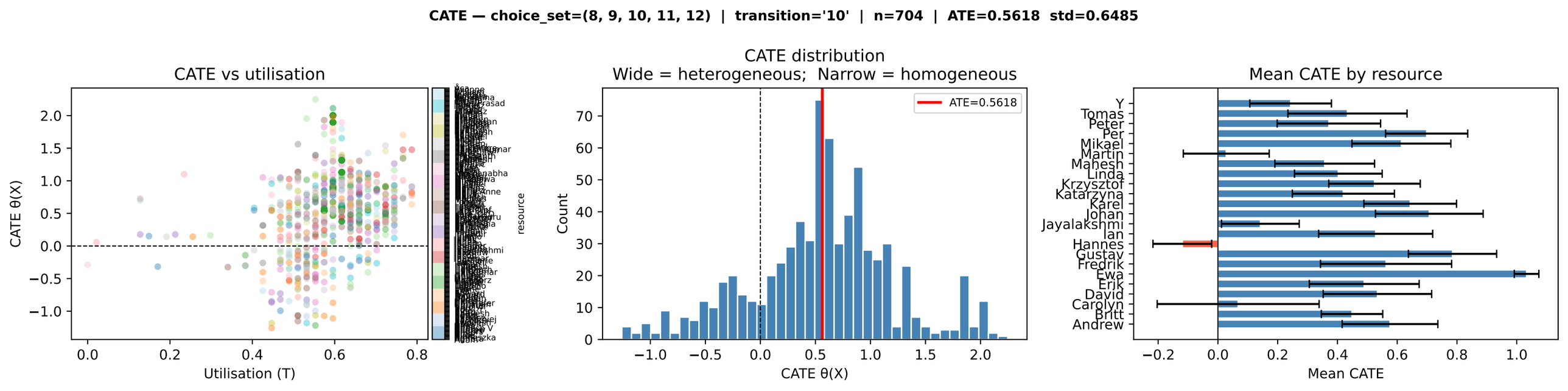

CATE summary tree | choice_set=(3, 4, 5, 6, 7) | transition='7' | top feature: alt_util_Awaiting Assignment_t3
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t3 <= 0.7257
samples = 3101
value = -2.6825
True
False
alt_util_Assigned_t5 <= 0.5393
samples = 2841
value = -2.9803
alt_util_In Progress_t4 <= 0.6925
samples = 260
value = 0.5715
alt_util_Awaiting Assignment_t3 <= 0.5868
samples = 745
value = -4.2042
alt_util_Closed_t6 <= 0.7155
samples = 2096
value = -2.5453
hour <= 10.5
samples = 224
value = 1.1734
hour <= 10.5
samples = 36
value = -3.1737
samples = 638
value = -3.5314
samples = 107
value = -8.2162
samples = 1291
value = -1.5929
samples = 805
value = -4.0726
samples = 93
value = 2.2408
samples = 131
value = 0.4156
samples = 16
value = -1.2023
samples = 20
value = -4.7508

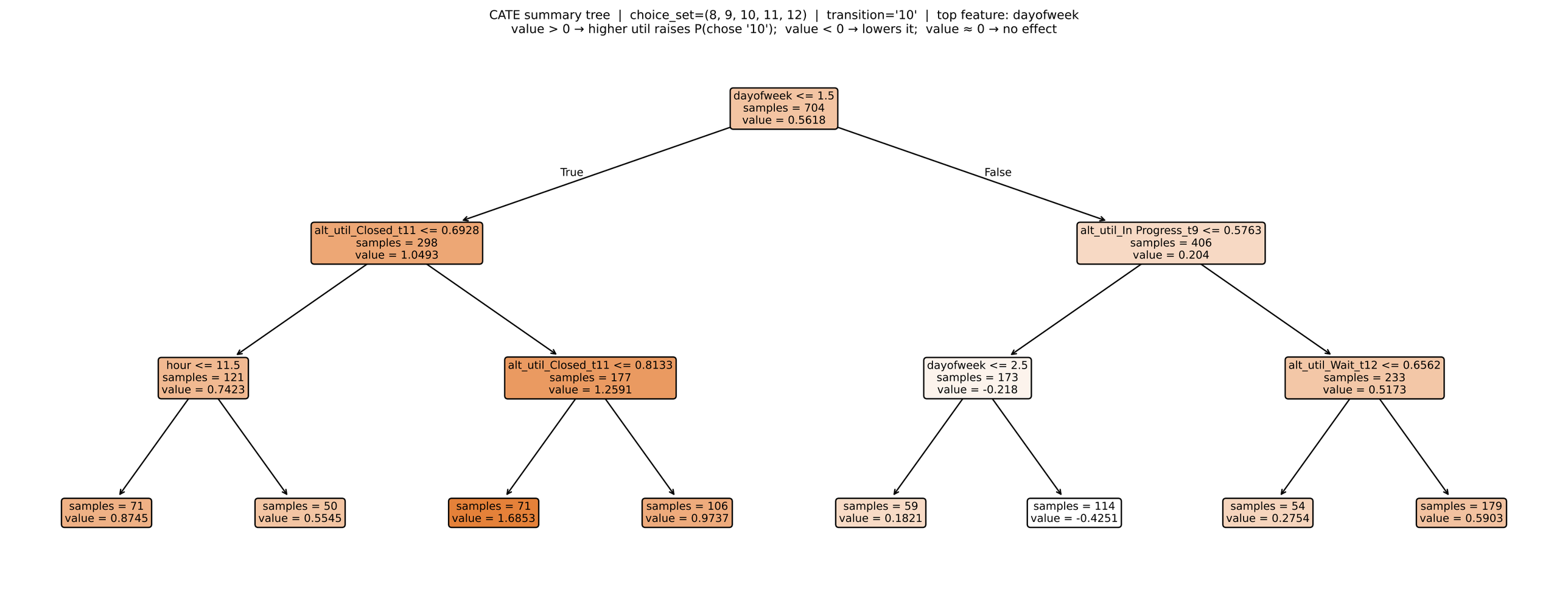

CATE — choice_set=(8, 9, 10, 11, 12) | transition='8' | n=704 | ATE=-0.0071 std=0.3138
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0071
Count
CATE θ(X)
Mean CATE by resource
Y
Tomas
Peter
Per
Mikael
Martin
Mahesh
Linda
Krzysztof
Katarzyna
Karel
Johan
Jayalakshmi
Ian
Hannes
Gustav
Fredrik
Ewa
Erik
David
Carolyn
Britt
Andrew
Mean CATE

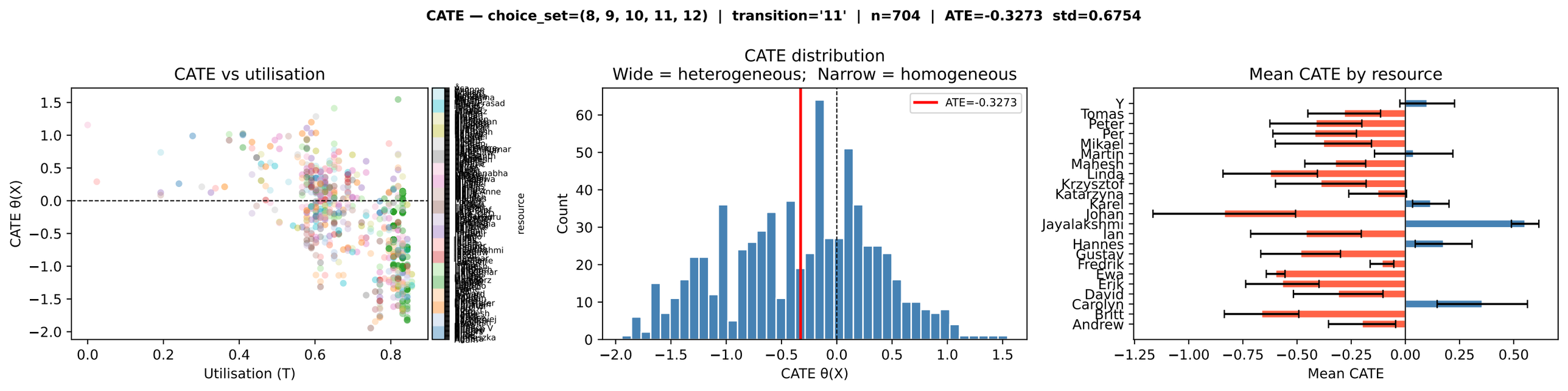

CATE summary tree | choice_set=(8, 9, 10, 11, 12) | transition='8' | top feature: alt_util_In Progress_t9
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_In Progress_t9 <= 0.5458
samples = 704
value = -0.0071
True
False
hour <= 11.5
samples = 199
value = 0.3846
dayofweek <= 2.5
samples = 505
value = -0.1615
alt_util_Closed_t11 <= 0.6446
samples = 106
value = 0.2869
alt_util_Wait_t12 <= 0.4479
samples = 93
value = 0.496
alt_util_In Progress_t9 <= 0.5687
samples = 328
value = -0.1028
dayofweek <= 3.5
samples = 177
value = -0.2705
samples = 75
value = 0.3431
samples = 31
value = 0.151
samples = 32
value = 0.3872
samples = 61
value = 0.5531
samples = 27
value = 0.1598
samples = 301
value = -0.1263
samples = 88
value = -0.3524
samples = 89
value = -0.1895

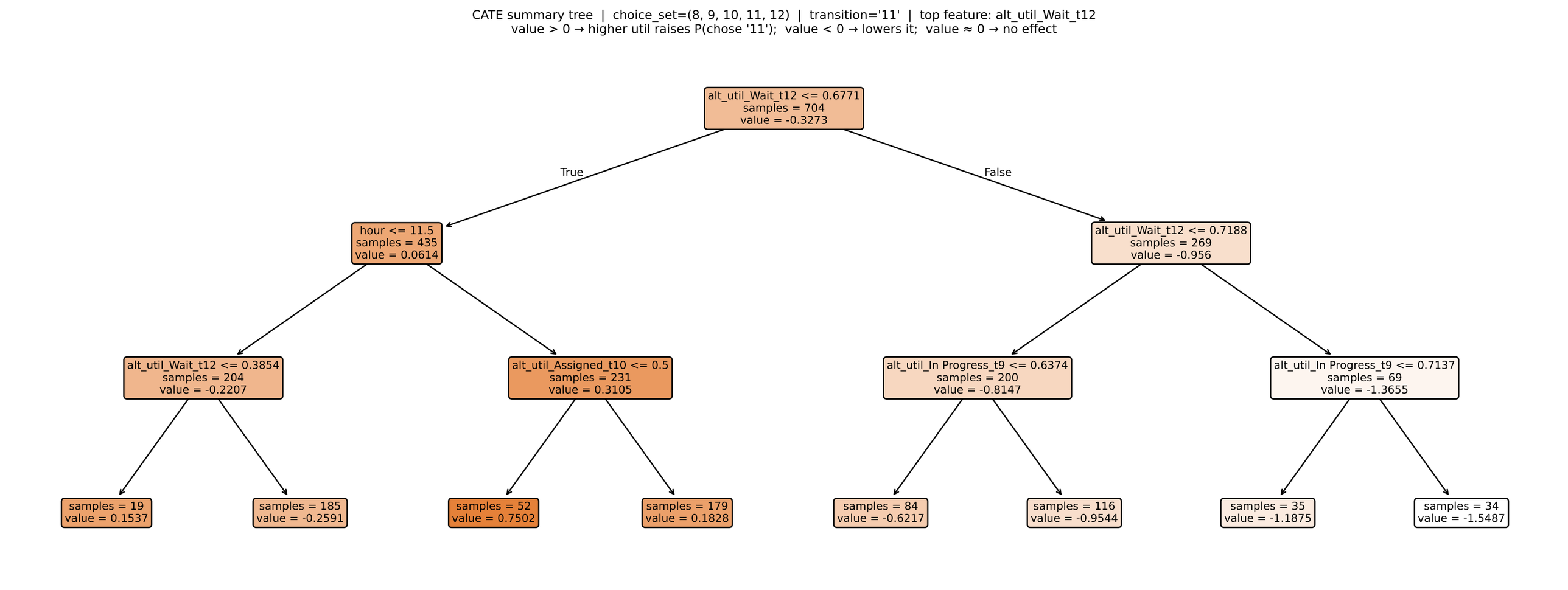

CATE — choice_set=(8, 9, 10, 11, 12) | transition='9' | n=704 | ATE=-1.8923 std=1.4445
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.8923
Count
CATE θ(X)
Mean CATE by resource
Y
Tomas
Peter
Per
Mikael
Martin
Mahesh
Linda
Krzysztof
Katarzyna
Karel
Johan
Jayalakshmi
Ian
Hannes
Gustav
Fredrik
Ewa
Erik
David
Carolyn
Britt
Andrew
Mean CATE

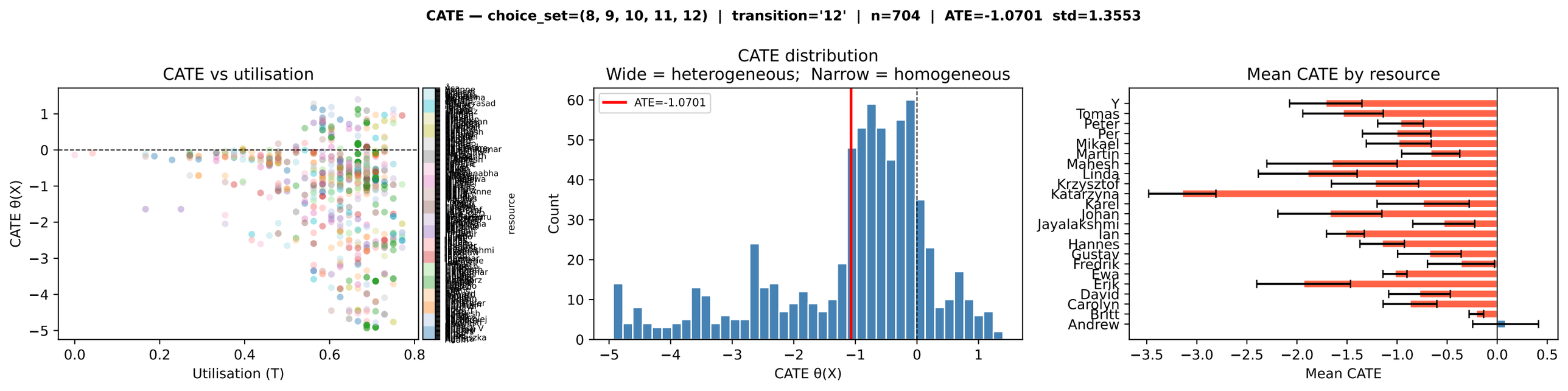

CATE summary tree | choice_set=(8, 9, 10, 11, 12) | transition='9' | top feature: alt_util_Assigned_t10
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t10 <= 0.7128
samples = 704
value = -1.8923
True
False
alt_util_Closed_t11 <= 0.8133
samples = 634
value = -1.5758
hour <= 11.5
samples = 70
value = -4.7583
alt_util_Wait_t12 <= 0.6979
samples = 460
value = -1.8824
dayofweek <= 0.5
samples = 174
value = -0.7653
res_Ewa <= 0.5
samples = 35
value = -4.0372
alt_util_Closed_t11 <= 0.6506
samples = 35
value = -5.4793
samples = 411
value = -1.7358
samples = 49
value = -3.1126
samples = 63
value = -1.7271
samples = 111
value = -0.2194
samples = 31
value = -4.212
samples = 4
value = -2.6828
samples = 1
value = -3.2343
samples = 34
value = -5.5453

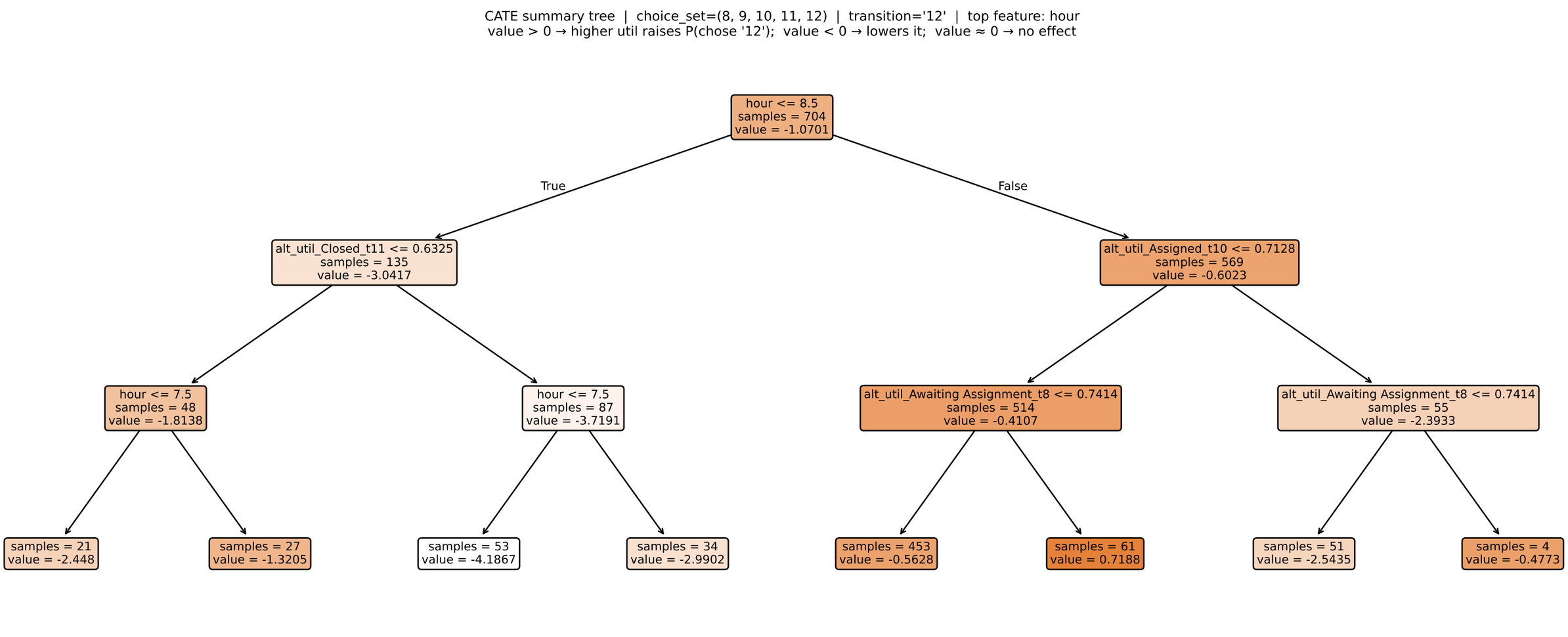

CATE — choice_set=(8, 9, 10, 11, 12) | transition='10' | n=704 | ATE=0.5618 std=0.6485
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.5618
Count
CATE θ(X)
Mean CATE by resource
Y
Tomas
Peter
Per
Mikael
Martin
Mahesh
Linda
Krzysztof
Katarzyna
Karel
Johan
Jayalakshmi
Ian
Hannes
Gustav
Fredrik
Ewa
Erik
David
Carolyn
Britt
Andrew
Mean CATE

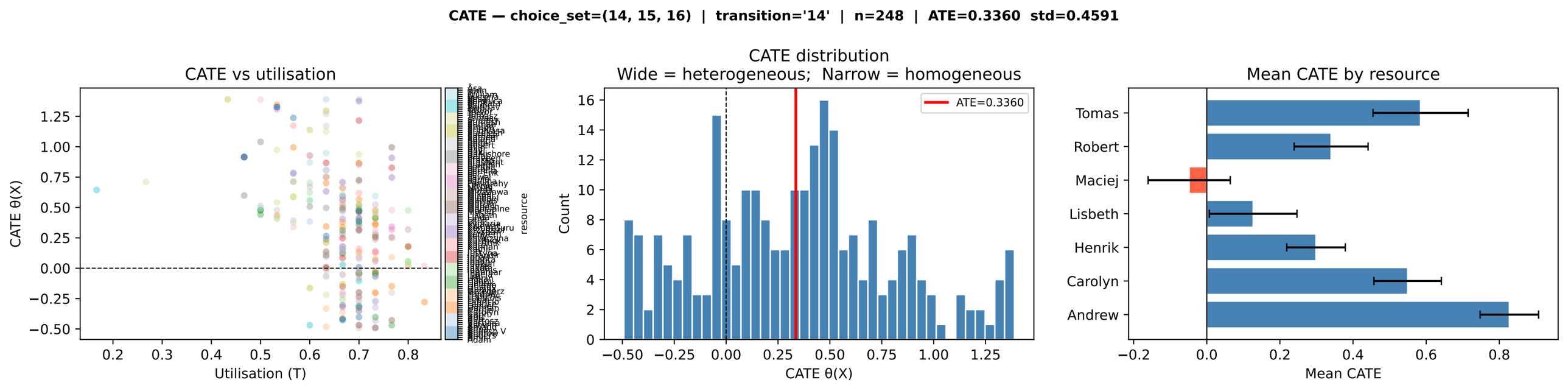

CATE summary tree | choice_set=(8, 9, 10, 11, 12) | transition='10' | top feature: dayofweek
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 1.5
samples = 704
value = 0.5618
True
False
alt_util_Closed_t11 <= 0.6928
samples = 298
value = 1.0493
alt_util_In Progress_t9 <= 0.5763
samples = 406
value = 0.204
hour <= 11.5
samples = 121
value = 0.7423
alt_util_Closed_t11 <= 0.8133
samples = 177
value = 1.2591
dayofweek <= 2.5
samples = 173
value = -0.218
alt_util_Wait_t12 <= 0.6562
samples = 233
value = 0.5173
samples = 71
value = 0.8745
samples = 50
value = 0.5545
samples = 71
value = 1.6853
samples = 106
value = 0.9737
samples = 59
value = 0.1821
samples = 114
value = -0.4251
samples = 54
value = 0.2754
samples = 179
value = 0.5903

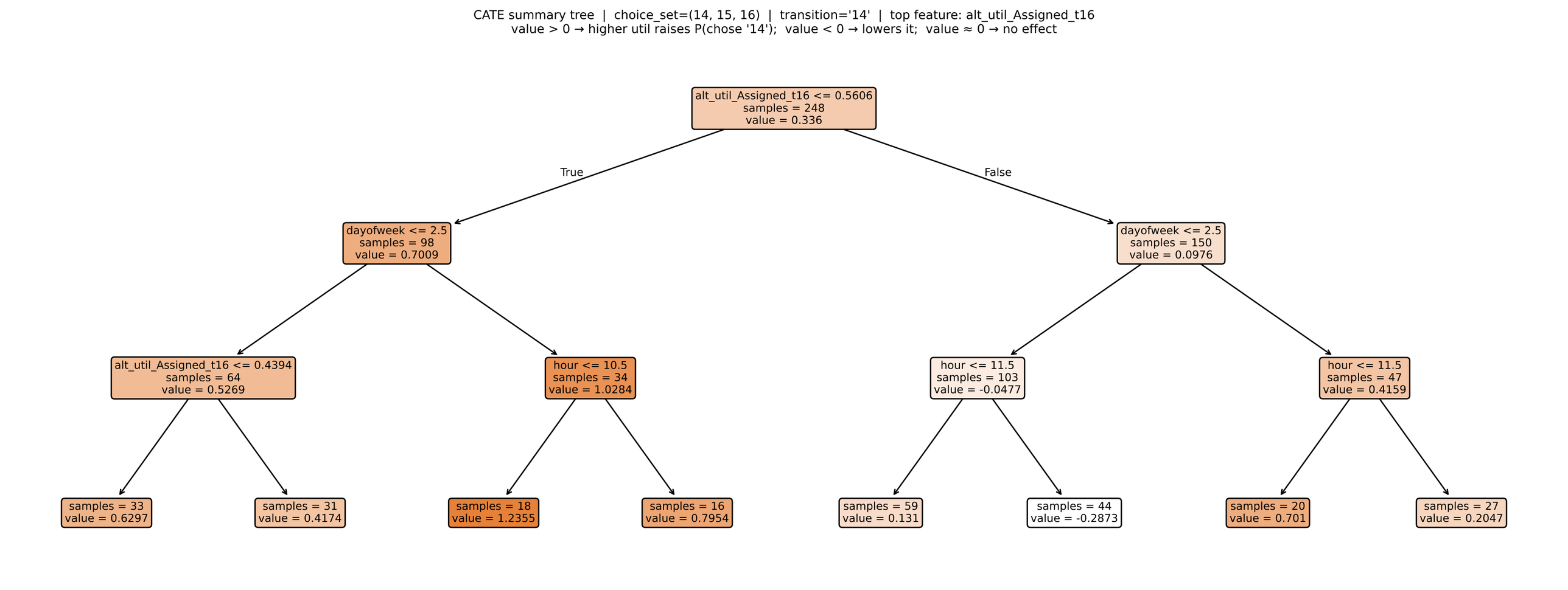

CATE — choice_set=(8, 9, 10, 11, 12) | transition='11' | n=704 | ATE=-0.3273 std=0.6754
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3273
Count
CATE θ(X)
Mean CATE by resource
Y
Tomas
Peter
Per
Mikael
Martin
Mahesh
Linda
Krzysztof
Katarzyna
Karel
Johan
Jayalakshmi
Ian
Hannes
Gustav
Fredrik
Ewa
Erik
David
Carolyn
Britt
Andrew
Mean CATE

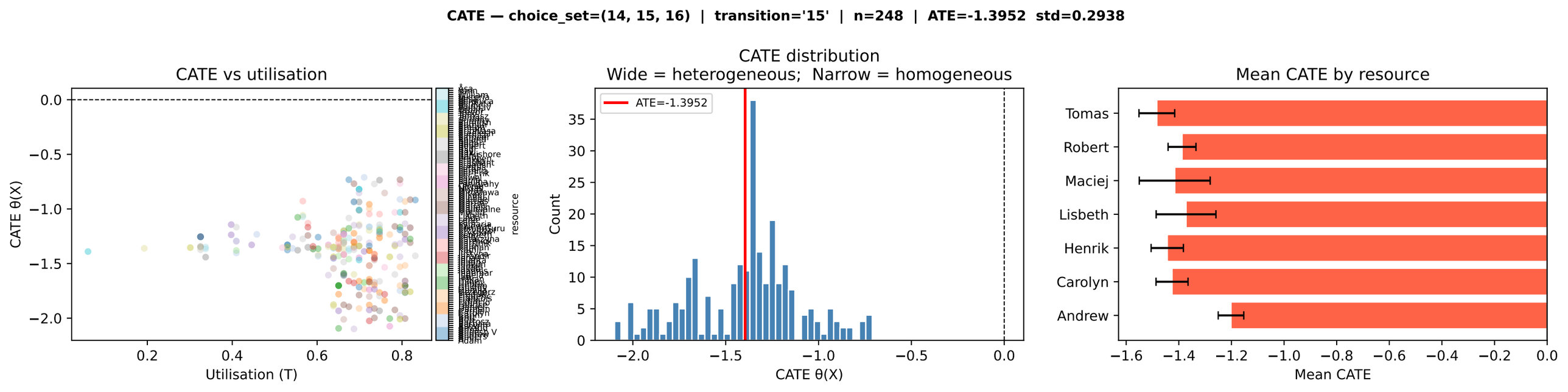

CATE summary tree | choice_set=(8, 9, 10, 11, 12) | transition='11' | top feature: alt_util_Wait_t12
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Wait_t12 <= 0.6771
samples = 704
value = -0.3273
True
False
hour <= 11.5
samples = 435
value = 0.0614
alt_util_Wait_t12 <= 0.7188
samples = 269
value = -0.956
alt_util_Wait_t12 <= 0.3854
samples = 204
value = -0.2207
alt_util_Assigned_t10 <= 0.5
samples = 231
value = 0.3105
alt_util_In Progress_t9 <= 0.6374
samples = 200
value = -0.8147
alt_util_In Progress_t9 <= 0.7137
samples = 69
value = -1.3655
samples = 19
value = 0.1537
samples = 185
value = -0.2591
samples = 52
value = 0.7502
samples = 179
value = 0.1828
samples = 84
value = -0.6217
samples = 116
value = -0.9544
samples = 35
value = -1.1875
samples = 34
value = -1.5487

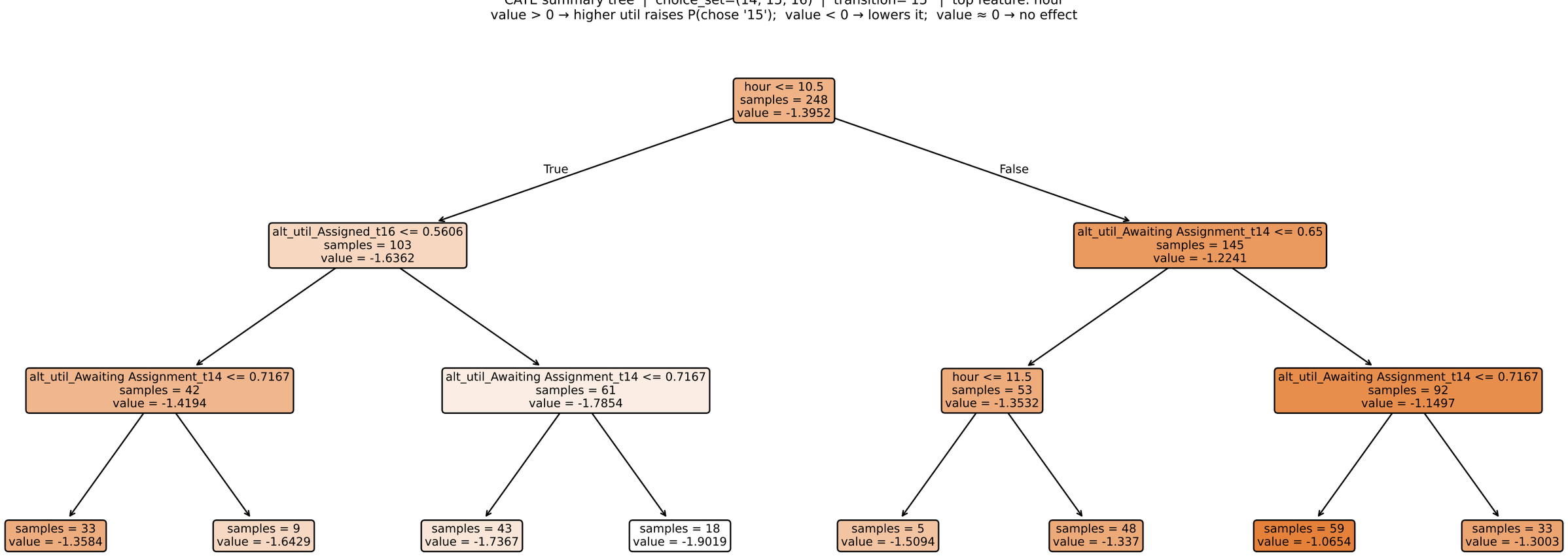

CATE — choice_set=(8, 9, 10, 11, 12) | transition='12' | n=704 | ATE=-1.0701 std=1.3553
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.0701
Count
CATE θ(X)
Mean CATE by resource
Y
Tomas
Peter
Per
Mikael
Martin
Mahesh
Linda
Krzysztof
Katarzyna
Karel
Johan
Jayalakshmi
Ian
Hannes
Gustav
Fredrik
Ewa
Erik
David
Carolyn
Britt
Andrew
Mean CATE

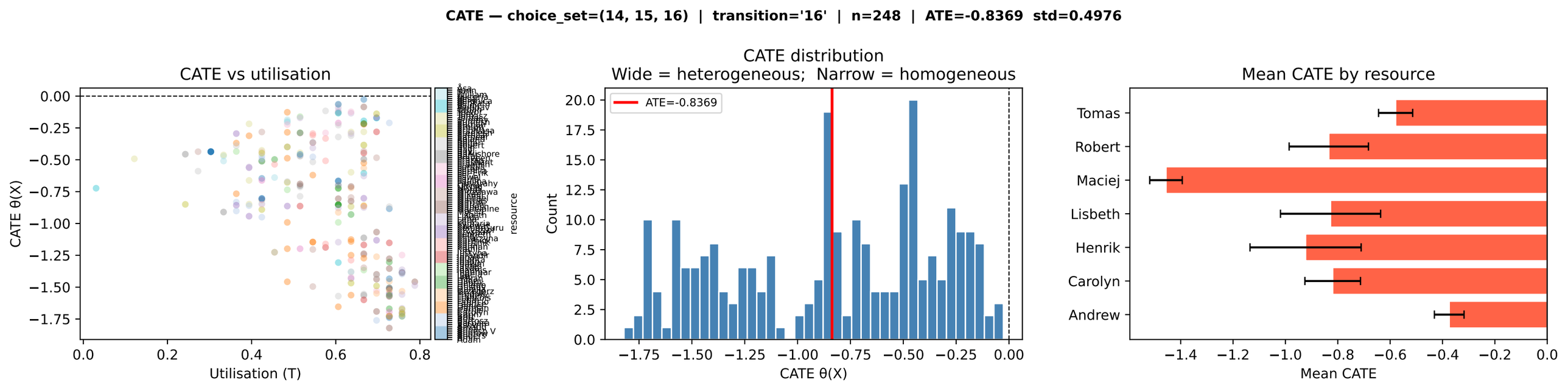

CATE summary tree | choice_set=(8, 9, 10, 11, 12) | transition='12' | top feature: hour
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 8.5
samples = 704
value = -1.0701
True
False
alt_util_Closed_t11 <= 0.6325
samples = 135
value = -3.0417
alt_util_Assigned_t10 <= 0.7128
samples = 569
value = -0.6023
hour <= 7.5
samples = 48
value = -1.8138
hour <= 7.5
samples = 87
value = -3.7191
alt_util_Awaiting Assignment_t8 <= 0.7414
samples = 514
value = -0.4107
alt_util_Awaiting Assignment_t8 <= 0.7414
samples = 55
value = -2.3933
samples = 21
value = -2.448
samples = 27
value = -1.3205
samples = 53
value = -4.1867
samples = 34
value = -2.9902
samples = 453
value = -0.5628
samples = 61
value = 0.7188
samples = 51
value = -2.5435
samples = 4
value = -0.4773

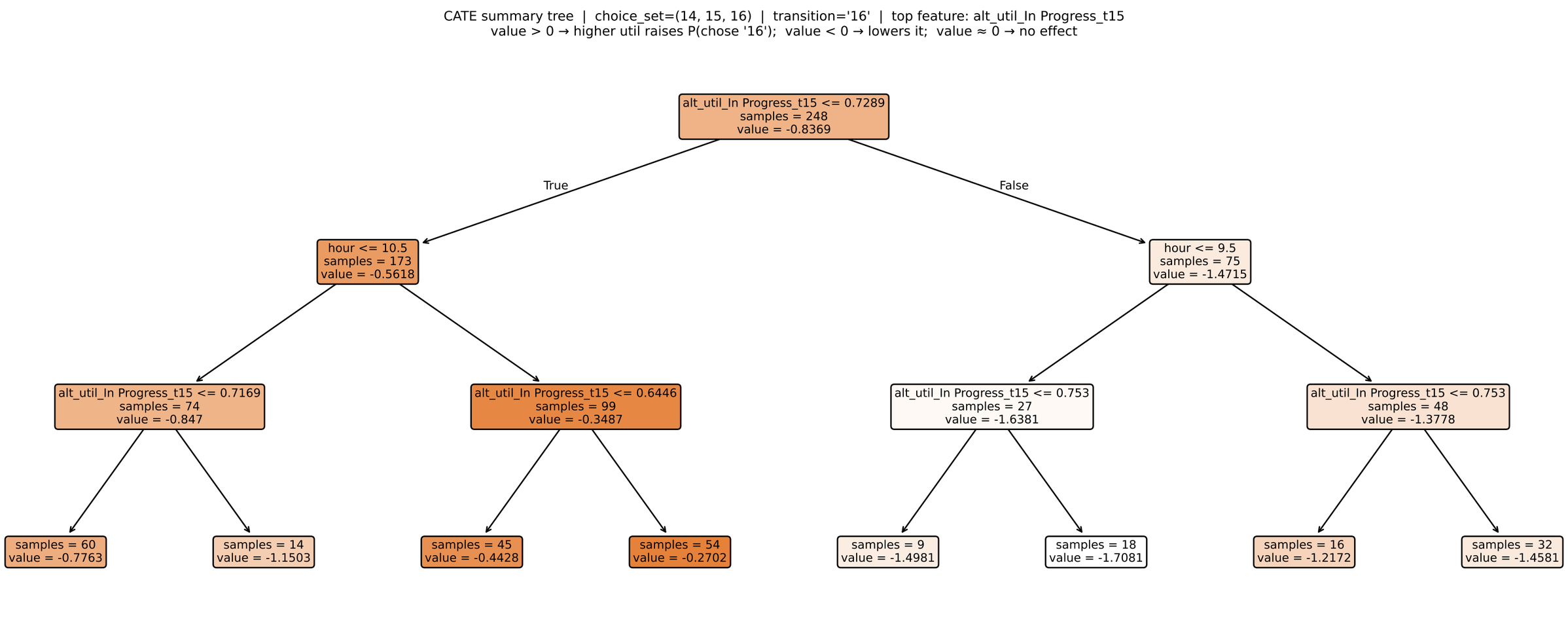

CATE — choice_set=(14, 15, 16) | transition='14' | n=248 | ATE=0.3360 std=0.4591
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.3360
Count
CATE θ(X)
Mean CATE by resource
Tomas
Robert
Maciej
Lisbeth
Henrik
Carolyn
Andrew
Mean CATE


CATE summary tree | choice_set=(14, 15, 16) | transition='14' | top feature: alt_util_Assigned_t16
value > 0 → higher util raises P(chose '14'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t16 <= 0.5606
samples = 248
value = 0.336
True
False
dayofweek <= 2.5
samples = 98
value = 0.7009
dayofweek <= 2.5
samples = 150
value = 0.0976
alt_util_Assigned_t16 <= 0.4394
samples = 64
value = 0.5269
hour <= 10.5
samples = 34
value = 1.0284
hour <= 11.5
samples = 103
value = -0.0477
hour <= 11.5
samples = 47
value = 0.4159
samples = 33
value = 0.6297
samples = 31
value = 0.4174
samples = 18
value = 1.2355
samples = 16
value = 0.7954
samples = 59
value = 0.131
samples = 44
value = -0.2873
samples = 20
value = 0.701
samples = 27
value = 0.2047

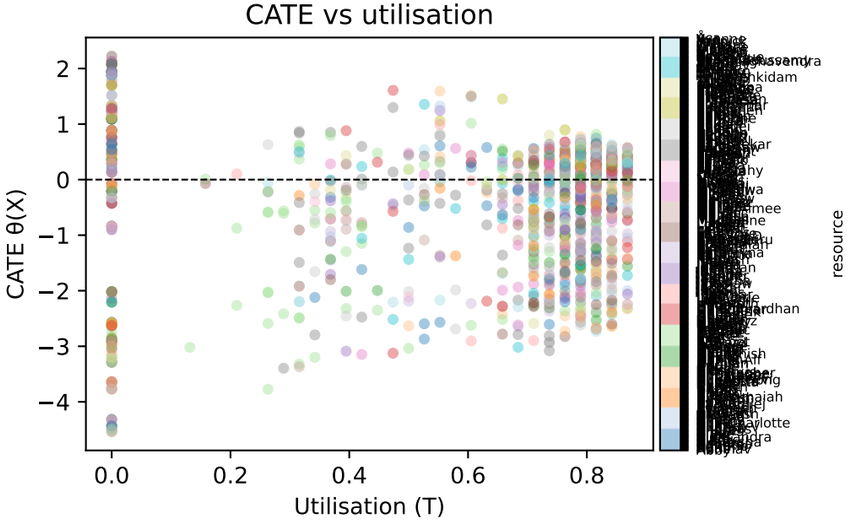

CATE — choice_set=(14, 15, 16) | transition='15' | n=248 | ATE=-1.3952 std=0.2938
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.3952
Count
CATE θ(X)
Mean CATE by resource
Tomas
Robert
Maciej
Lisbeth
Henrik
Carolyn
Andrew
Mean CATE

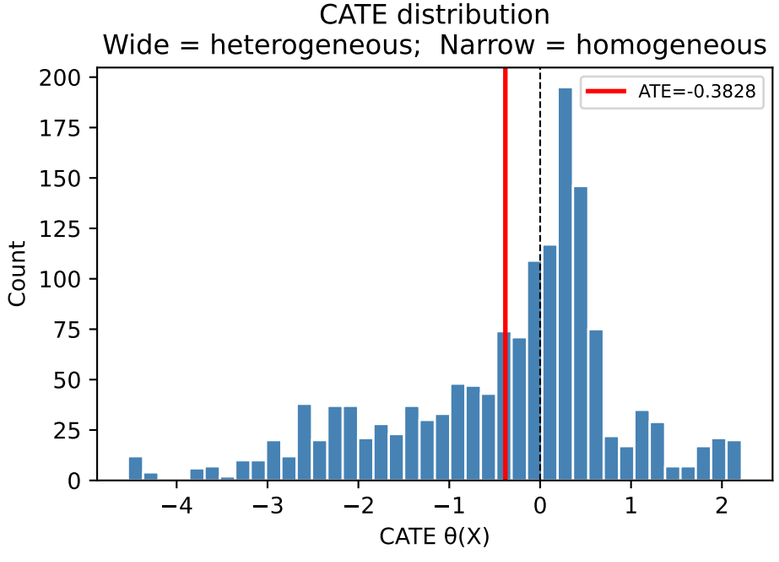

CATE summary tree | choice_set=(14, 15, 16) | transition='15' | top feature: hour
value > 0 → higher util raises P(chose '15'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 10.5
samples = 248
value = -1.3952
True
False
alt_util_Assigned_t16 <= 0.5606
samples = 103
value = -1.6362
alt_util_Awaiting Assignment_t14 <= 0.65
samples = 145
value = -1.2241
alt_util_Awaiting Assignment_t14 <= 0.7167
samples = 42
value = -1.4194
alt_util_Awaiting Assignment_t14 <= 0.7167
samples = 61
value = -1.7854
hour <= 11.5
samples = 53
value = -1.3532
alt_util_Awaiting Assignment_t14 <= 0.7167
samples = 92
value = -1.1497
samples = 33
value = -1.3584
samples = 9
value = -1.6429
samples = 43
value = -1.7367
samples = 18
value = -1.9019
samples = 5
value = -1.5094
samples = 48
value = -1.337
samples = 59
value = -1.0654
samples = 33
value = -1.3003

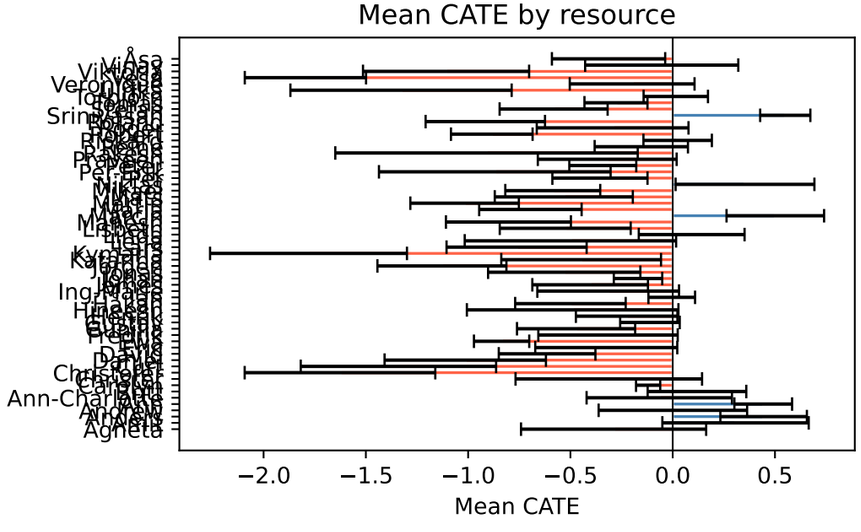

CATE — choice_set=(14, 15, 16) | transition='16' | n=248 | ATE=-0.8369 std=0.4976
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.8369
Count
CATE θ(X)
Mean CATE by resource
Tomas
Robert
Maciej
Lisbeth
Henrik
Carolyn
Andrew
Mean CATE

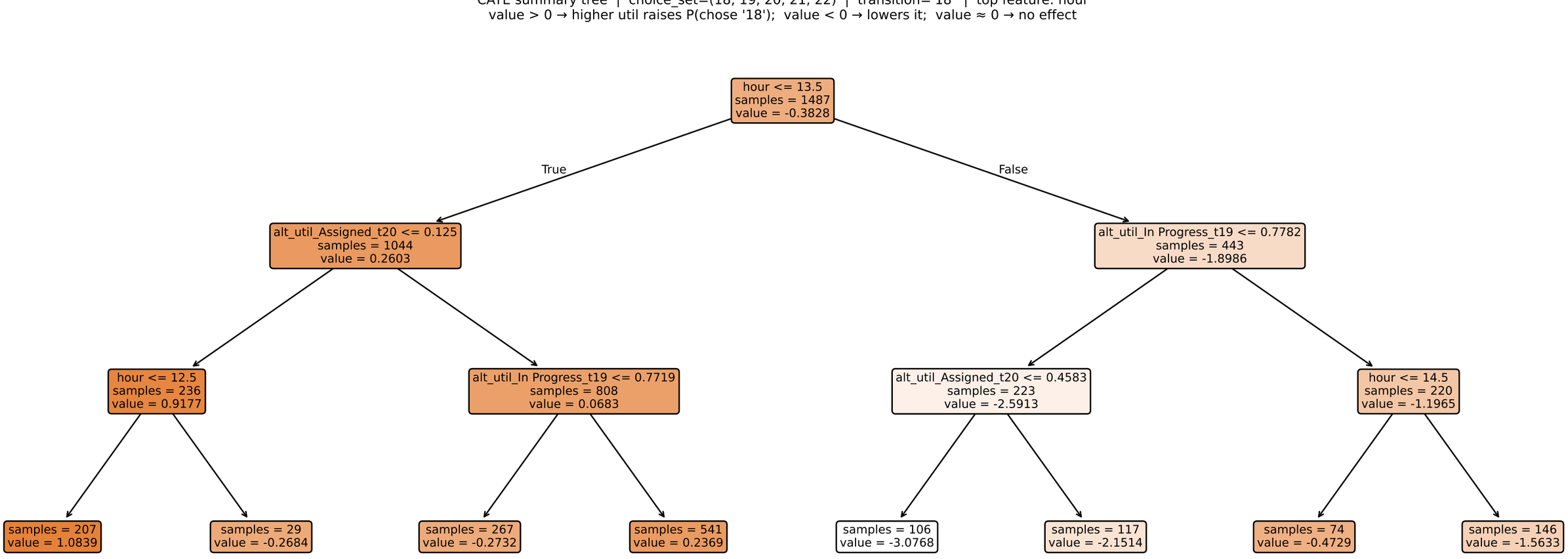

CATE summary tree | choice_set=(14, 15, 16) | transition='16' | top feature: alt_util_ln Progress_t15
value > 0 → higher util raises P(chose '16'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t15 <= 0.7289
samples = 248
value = -0.8369
True
False
hour <= 10.5
samples = 173
value = -0.5618
hour <= 9.5
samples = 75
value = -1.4715
alt_util_ln Progress_t15 <= 0.7169
samples = 74
value = -0.847
alt_util_ln Progress_t15 <= 0.6446
samples = 99
value = -0.3487
alt_util_ln Progress_t15 <= 0.753
samples = 27
value = -1.6381
alt_util_ln Progress_t15 <= 0.753
samples = 48
value = -1.3778
samples = 60
value = -0.7763
samples = 14
value = -1.1503
samples = 45
value = -0.4428
samples = 54
value = -0.2702
samples = 9
value = -1.4981
samples = 18
value = -1.7081
samples = 16
value = -1.2172
samples = 32
value = -1.4581

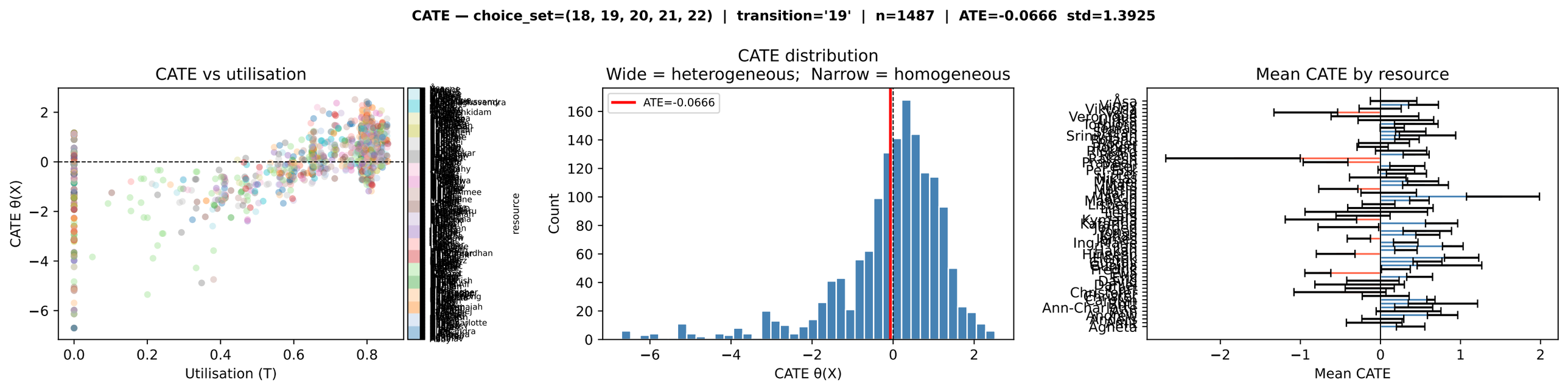
CATE — choice_set=(18, 19, 20, 21, 22) | transition='18' | n=1487 | ATE=-0.3828 std=1.2711

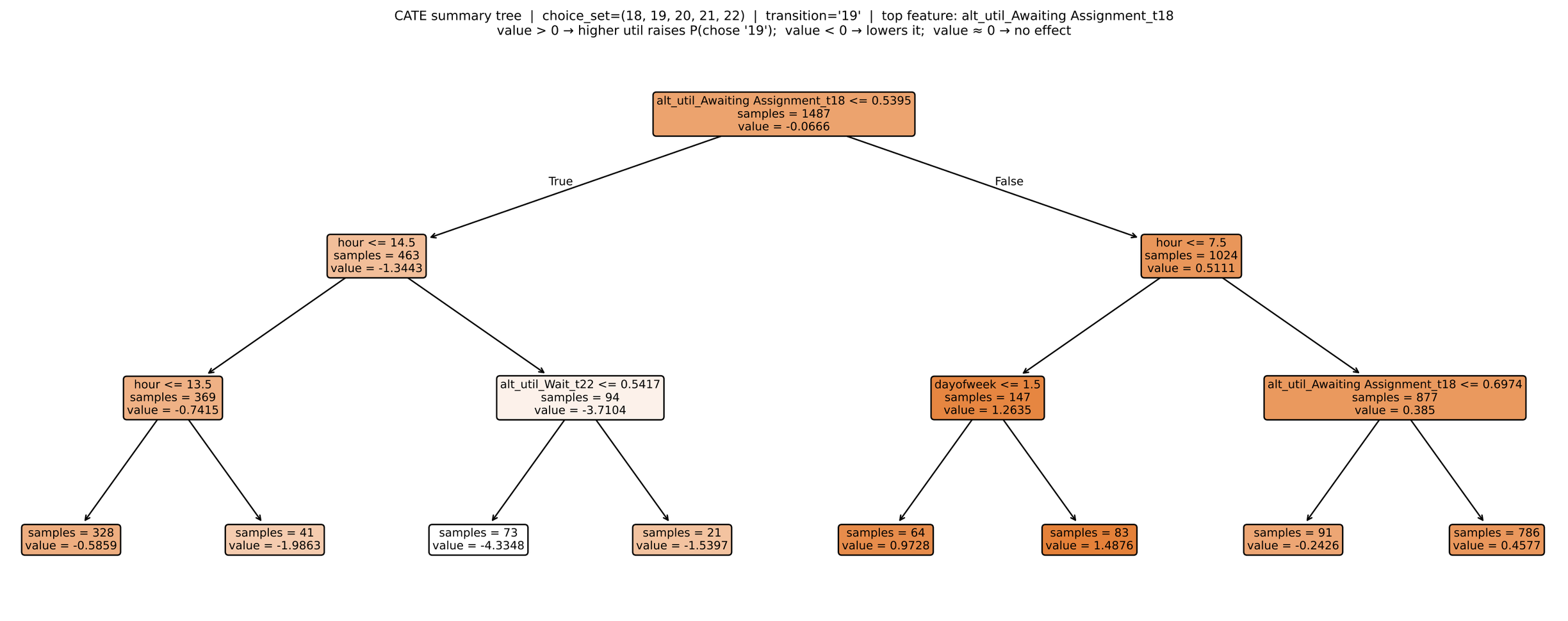


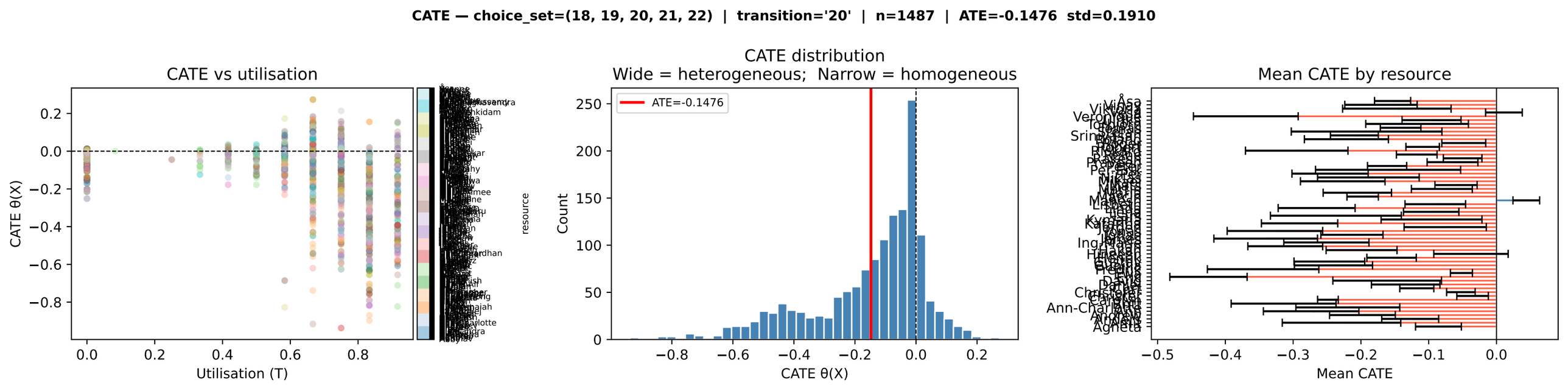


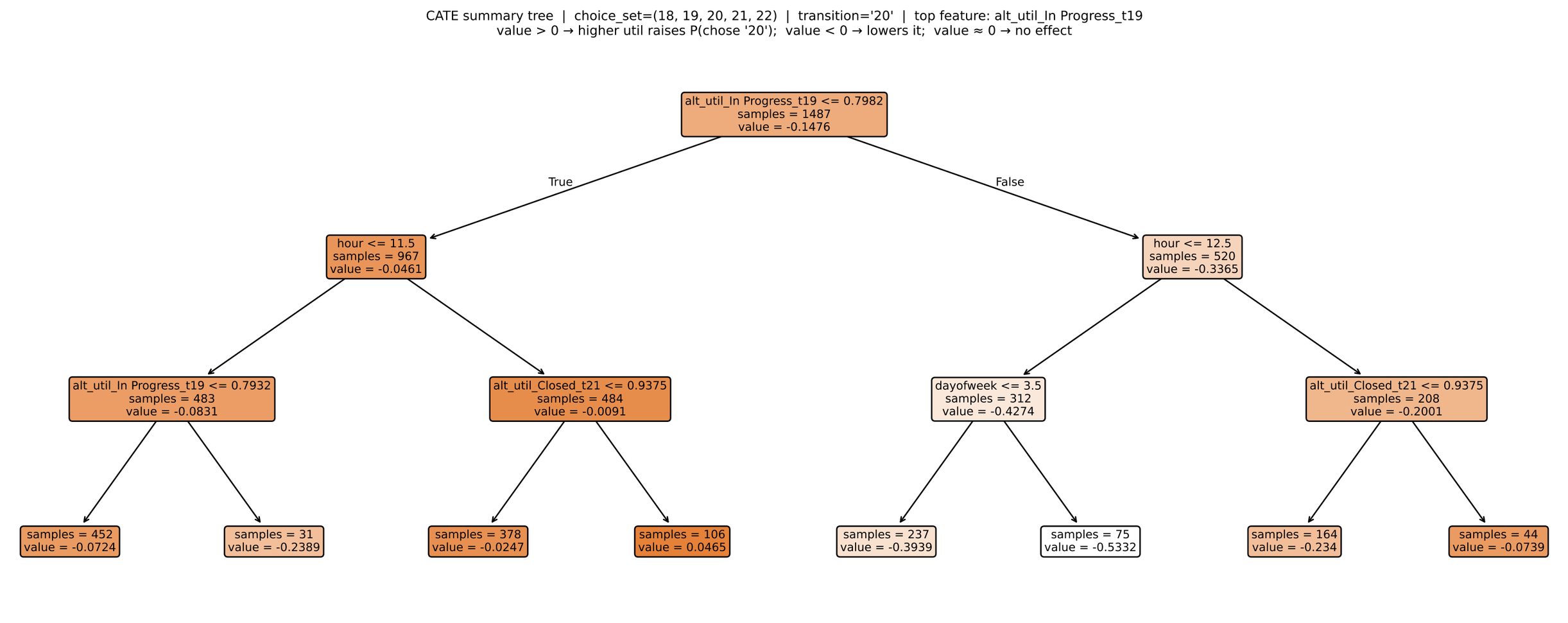


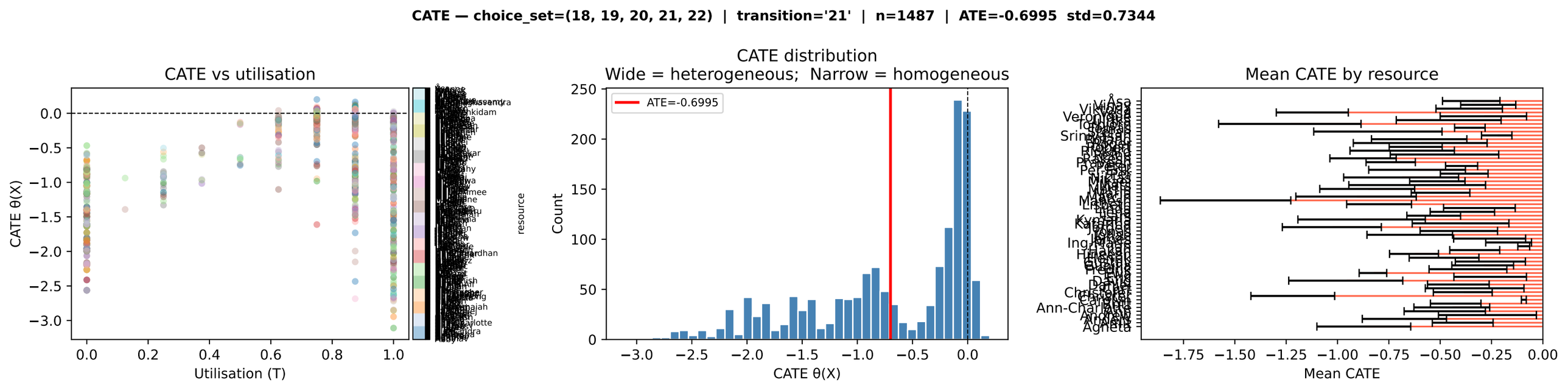

CATE summary tree | choice_set=(18, 19, 20, 21, 22) | transition='18' | top feature: hour
value > 0 → higher util raises P(chose '18'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 13.5
samples = 1487
value = -0.3828
True
False
alt_util_Assigned_t20 <= 0.125
samples = 1044
value = 0.2603
alt_util_In Progress_t19 <= 0.7782
samples = 443
value = -1.8986
hour <= 12.5
samples = 236
value = 0.9177
alt_util_In Progress_t19 <= 0.7719
samples = 808
value = 0.0683
alt_util_Assigned_t20 <= 0.4583
samples = 223
value = -2.5913
hour <= 14.5
samples = 220
value = -1.1965
samples = 207
value = 1.0839
samples = 29
value = -0.2684
samples = 267
value = -0.2732
samples = 541
value = 0.2369
samples = 106
value = -3.0768
samples = 117
value = -2.1514
samples = 74
value = -0.4729
samples = 146
value = -1.5633

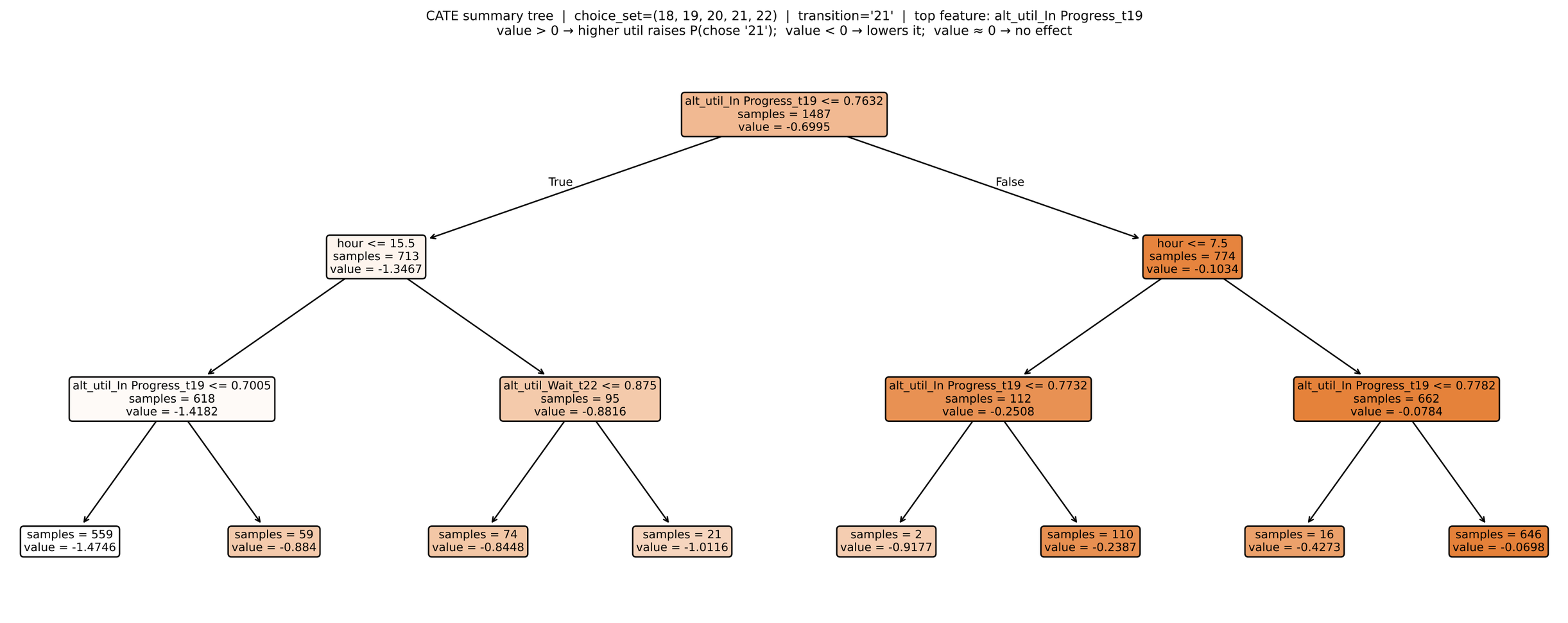

CATE — choice_set=(18, 19, 20, 21, 22) | transition='19' | n=1487 | ATE=-0.0666 std=1.3925
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0666
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

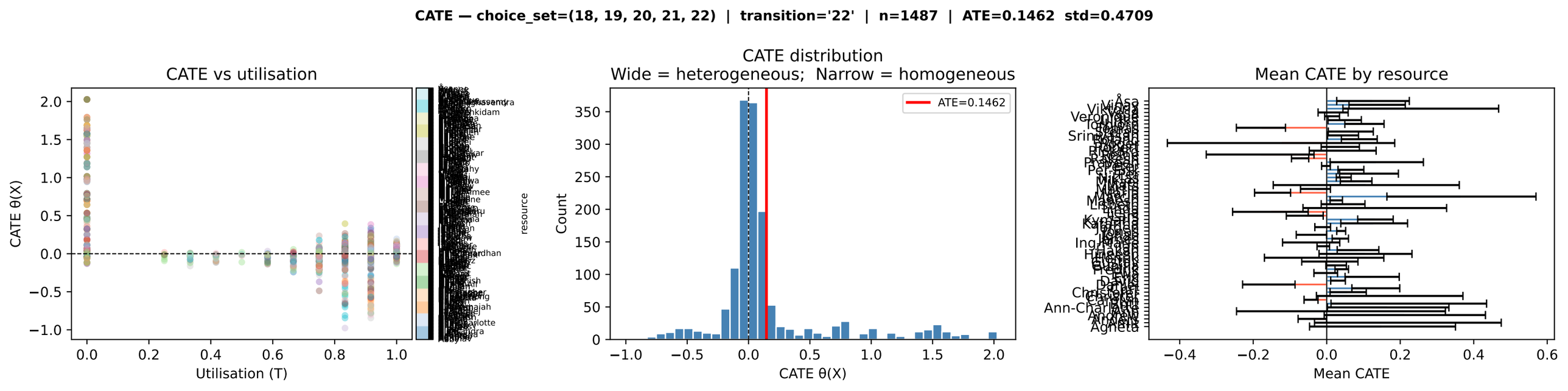

CATE summary tree | choice_set=(18, 19, 20, 21, 22) | transition='19' | top feature: alt_util_Awaiting Assignment_t18
value > 0 → higher util raises P(chose '19'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t18 <= 0.5395
samples = 1487
value = -0.0666
True
False
hour <= 14.5
samples = 463
value = -1.3443
hour <= 7.5
samples = 1024
value = 0.5111
hour <= 13.5
samples = 369
value = -0.7415
alt_util_Wait_t22 <= 0.5417
samples = 94
value = -3.7104
dayofweek <= 1.5
samples = 147
value = 1.2635
alt_util_Awaiting Assignment_t18 <= 0.6974
samples = 877
value = 0.385
samples = 328
value = -0.5859
samples = 41
value = -1.9863
samples = 73
value = -4.3348
samples = 21
value = -1.5397
samples = 64
value = 0.9728
samples = 83
value = 1.4876
samples = 91
value = -0.2426
samples = 786
value = 0.4577

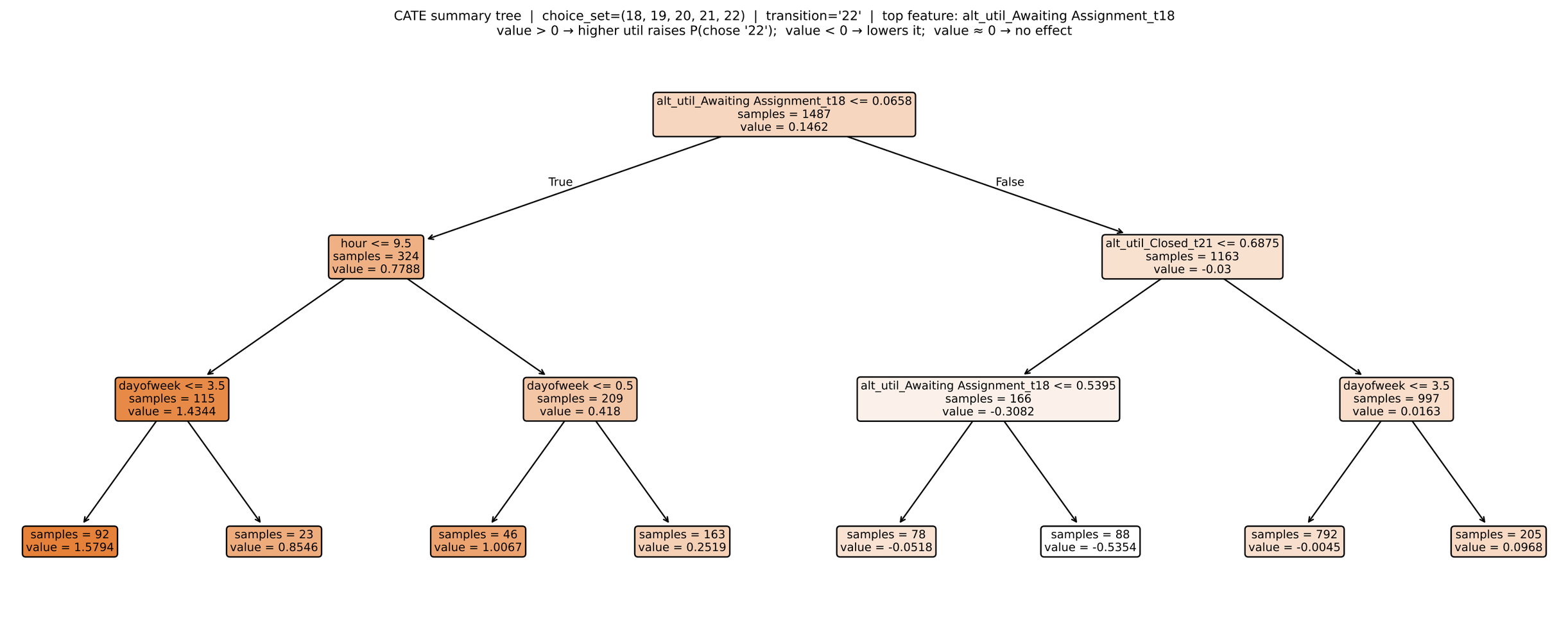

CATE — choice_set=(18, 19, 20, 21, 22) | transition='20' | n=1487 | ATE=-0.1476 std=0.1910
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1476
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(18, 19, 20, 21, 22) | transition='20' | top feature: alt_util_ln Progress_t19
value > 0 → higher util raises P(chose '20'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t19 <= 0.7982
samples = 1487
value = -0.1476
True
False
hour <= 11.5
samples = 967
value = -0.0461
hour <= 12.5
samples = 520
value = -0.3365
alt_util_ln Progress_t19 <= 0.7932
samples = 483
value = -0.0831
alt_util_Closed_t21 <= 0.9375
samples = 484
value = -0.0091
dayofweek <= 3.5
samples = 312
value = -0.4274
alt_util_Closed_t21 <= 0.9375
samples = 208
value = -0.2001
samples = 452
value = -0.0724
samples = 31
value = -0.2389
samples = 378
value = -0.0247
samples = 106
value = 0.0465
samples = 237
value = -0.3939
samples = 75
value = -0.5332
samples = 164
value = -0.234
samples = 44
value = -0.0739


CATE — choice_set=(18, 19, 20, 21, 22) | transition='21' | n=1487 | ATE=-0.6995 std=0.7344
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.6995
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(18, 19, 20, 21, 22) | transition='21' | top feature: alt_util_ln Progress_t19
value > 0 → higher util raises P(chose '21'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t19 <= 0.7632
samples = 1487
value = -0.6995
True
False
hour <= 15.5
samples = 713
value = -1.3467
hour <= 7.5
samples = 774
value = -0.1034
alt_util_ln Progress_t19 <= 0.7005
samples = 618
value = -1.4182
alt_util_Wait_t22 <= 0.875
samples = 95
value = -0.8816
alt_util_ln Progress_t19 <= 0.7732
samples = 112
value = -0.2508
alt_util_ln Progress_t19 <= 0.7782
samples = 662
value = -0.0784
samples = 559
value = -1.4746
samples = 59
value = -0.884
samples = 74
value = -0.8448
samples = 21
value = -1.0116
samples = 2
value = -0.9177
samples = 110
value = -0.2387
samples = 16
value = -0.4273
samples = 646
value = -0.0698


CATE — choice_set=(18, 19, 20, 21, 22) | transition='22' | n=1487 | ATE=0.1462 std=0.4709
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.1462
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(18, 19, 20, 21, 22) | transition='22' | top feature: alt_util_Awaiting Assignment_t18
value > 0 → higher util raises P(chose '22'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t18 <= 0.0658
samples = 1487
value = 0.1462
True
False
hour <= 9.5
samples = 324
value = 0.7788
alt_util_Closed_t21 <= 0.6875
samples = 1163
value = -0.03
dayofweek <= 3.5
samples = 115
value = 1.4344
dayofweek <= 0.5
samples = 209
value = 0.418
alt_util_Awaiting Assignment_t18 <= 0.5395
samples = 166
value = -0.3082
dayofweek <= 3.5
samples = 997
value = 0.0163
samples = 92
value = 1.5794
samples = 23
value = 0.8546
samples = 46
value = 1.0067
samples = 163
value = 0.2519
samples = 78
value = -0.0518
samples = 88
value = -0.5354
samples = 792
value = -0.0045
samples = 205
value = 0.0968

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(0, 1, 2)	0	875	0.251788	0.562488	-1.06112	-0.163757	0.164769	0.633605	1.704562	64.3	35.7	alt_util_In Progress_t1
(0, 1, 2)	1	875	0.678355	0.683115	-0.913032	0.173649	0.601411	1.094641	2.82957	85.3	14.7	dayofweek
(0, 1, 2)	2	875	-0.006889	0.461169	-1.598341	-0.271388	0.04022	0.296878	0.987367	53.4	46.6	alt_util_Awaiting Assignment_t0
(3, 4, 5, 6, 7)	3	3101	-1.662155	2.495484	-11.607017	-3.366432	-1.342127	0.104212	5.271107	26.8	73.2	alt_util_Closed_t6
(3, 4, 5, 6, 7)	4	3101	-4.232796	3.321176	-17.259049	-6.131806	-4.050646	-1.871936	4.263508	8.9	91.1	alt_util_Closed_t6
(3, 4, 5, 6, 7)	5	3101	-2.176946	2.530886	-10.633598	-3.773225	-1.906921	-0.510284	6.446913	17.8	82.2	alt_util_Closed_t6
(3, 4, 5, 6, 7)	6	3101	-1.904017	3.312985	-13.882547	-3.880604	-1.774226	0.246983	12.403338	27.7	72.3	hour
(3, 4, 5, 6, 7)	7	3101	-2.682526	2.634364	-13.522011	-4.095349	-2.657349	-1.028323	5.823901	14.6	85.4	alt_util_Awaiting Assignment_t3
(8, 9, 10, 11, 12)	8	704	-0.007146	0.313815	-0.91712	-0.222151	-0.087601	0.226114	0.938965	38.9	61.1	alt_util_In Progress_t9
(8, 9, 10, 11, 12)	9	704	-1.892258	1.444487	-6.907197	-2.465968	-1.739992	-1.10278	1.191467	9.4	90.6	alt_util_Assigned_t10
(8, 9, 10, 11, 12)	10	704	0.561794	0.648487	-1.258842	0.222596	0.585458	0.92198	2.247116	81.8	18.2	dayofweek
(8, 9, 10, 11, 12)	11	704	-0.327334	0.675426	-1.944126	-0.831844	-0.2335	0.159366	1.54612	35.7	64.3	alt_util_Wait_t12
(8, 9, 10, 11, 12)	12	704	-1.070115	1.355329	-4.931519	-1.639222	-0.745972	-0.178657	1.398587	15.8	84.2	hour
(14, 15, 16)	14	248	0.335976	0.459107	-0.492586	-0.014988	0.34988	0.600666	1.389057	74.2	25.8	alt_util_Assigned_t16
(14, 15, 16)	15	248	-1.395221	0.293757	-2.098373	-1.607007	-1.353357	-1.232255	-0.710716	0.0	100.0	hour
(14, 15, 16)	16	248	-0.836938	0.497591	-1.821173	-1.279996	-0.791516	-0.436363	-0.02646	0.0	100.0	alt_util_In Progress_t15
(18, 19, 20, 21, 22)	18	1487	-0.382844	1.271101	-4.537017	-1.076898	-0.025719	0.378671	2.219368	48.8	51.2	hour
(18, 19, 20, 21, 22)	19	1487	-0.066567	1.392484	-6.704065	-0.476405	0.216242	0.78886	2.516185	59.3	40.7	alt_util_Awaiting Assignment_t18
(18, 19, 20, 21, 22)	20	1487	-0.147636	0.191003	-0.937457	-0.228886	-0.085378	-0.013157	0.274168	16.7	83.3	alt_util_In Progress_t19
(18, 19, 20, 21, 22)	21	1487	-0.699531	0.73445	-3.109657	-1.14952	-0.356583	-0.072668	0.200158	8.6	91.4	alt_util_In Progress_t19
(18, 19, 20, 21, 22)	22	1487	0.146208	0.470892	-0.977248	-0.040875	0.016919	0.11666	2.02704	58.0	42.0	alt_util_Awaiting Assignment_t18

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md